



无序固体中的单极子屏蔽效应

学 生： 李梓瑞

指导教师： 白玛窦、崔健

摘 要

在无序固体中，塑性响应与应力屏蔽效应在非热体系的反常力学行为中起着核心作用。已有研究提出四极子与偶极子屏蔽机制，用于描述准弹性相与塑性固体相，但未堵塞区域中的力学响应机制仍缺乏清晰理解。本文基于反常弹性的几何理论框架，研究了无序固体中的单极子屏蔽效应。通过引入四极子、偶极子与单极子构成的屏蔽层级结构，我们推导得到平面几何下描述位移场的单极子屏蔽方程。在径向对称条件下，该方程可化为双亥姆霍兹方程，其解析解能够表示为 Kelvin 函数形式。所得位移场呈现振荡指数衰减行为，并由新的屏蔽长度尺度所刻画，这与偶极子屏蔽对应的弱屏蔽响应形成显著区别。为研究单极子屏蔽的物理效应，本文利用 LAMMPS 软件的数据，其在二维环形颗粒系统中采用非热准静态膨胀过程进行了分子动力学模拟。通过对预堵塞区域径向位移场的拟合分析，我们发现由几何理论导出的单极子屏蔽解能够较好地符合数值结果，尤其是在远离堵塞点的低体积分数区域。随着体系接近堵塞转变，位移场中逐渐出现振荡结构，而由偶极子与单极子组成的唯象联合屏蔽模型能够较好地刻画该行为，表明接近堵塞点时存在弹性类偶极传播与强屏蔽单极响应之间的共存与竞争。此外，本文定义并比较了相互作用长度尺度、关联长度尺度以及衰减长度尺度三种特征尺度，并发现它们在堵塞点附近表现出近似一致的标度行为。本文结果支持了单极子屏蔽主导预堵塞相力学响应的物理图像，并为类流体的非堵塞状态与堵塞塑性固体状态之间的连续过渡提供了一种连续介质理论描述。

关键词：无序固体，反常弹性，塑性，单极子屏蔽，颗粒材料



Effects of monopole screening in disordered solids

Author: Zirui Li

Tutor: Matteo Baggioli, Jian Cui

Abstract

In amorphous solids, plastic responses and stress-screening effects play central roles in anomalous mechanical behaviors for athermal systems. While quadrupole and dipole screening mechanisms have been proposed previously to describe quasielastic and plastic-solid phases, the mechanical response in the unjammed regime remains poorly understood. In this work, we investigate monopole screening effects in disordered solids within the framework of the geometric theory of anomalous elasticity. Starting from the hierarchy of quadrupolar, dipolar and monopolar screening modes, we derive a monopole screening equation governing the displacement field in planar geometry. Under radial symmetry, the equation reduces to a bi-Helmholtz equation with analytical solutions expressed in terms of Kelvin functions. The resulting displacement fields exhibit oscillatory exponentially decaying behavior characterized by new screening length scales, in contrast to the weakly screened dipole response. To examine the physical consequences of monopole screening, we use data by the LAMMPS package under athermal quasistatic inflation protocols, which performs molecular-dynamics simulations of two-dimensional granular systems in annular geometry. By fitting radial displacement fields in the pre-jamming regime, we show that monopole-screening solutions derived from the geometric theory are in agreement with numerical data, especially far below the jamming point. Near the jamming transition, oscillatory displacement profiles emerge and are accurately captured by a phenomenological combined dipole-monopole screening ansatz, suggesting coexistence and competition between elastic-like dipole propagation and strongly screened monopole responses. Furthermore, we define and compare three characteristic length scales, namely the interaction length scale, correlation length scale and decay length scale, and demonstrate their approximately identical scaling behavior near the jamming point. Our results support the interpretation



that monopole screening dominates the mechanical response in the pre-jamming phase and provide a continuum description for the crossover between fluid-like unjammed states and jammed plastic solid states.

Key words: Disordered solid, Anomalous elasticity, Plasticity, Monopole screening, Granular materials



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1 Introduction

1.1 Elasticity

Elasticity theory is a central part of soft matter physics and condensed matter physics, and it provides the basic language for describing the mechanical response of solid materials. As a continuum theoretical framework, it studies the deformation and related mechanical responses of ordered solids, namely crystals, under external loads. In this description, the microscopic positions of all atoms are not followed one by one. Instead, the material is represented through continuous fields, so that its macroscopic deformation can be treated in a systematic way. The development of elasticity theory has a long history, with its origins traceable to Hooke's law in the 17th century. Hooke's law revealed the linear relation between force and deformation under small-strain conditions, and thereby supplied the elementary physical picture behind linear elastic response. In the early 19th century, Cauchy formally introduced the concepts of stress and strain, and established the geometric equations, equilibrium differential equations, and generalized Hooke's law of elastic mechanics. These developments turned elasticity from an empirical rule into a coherent continuum theory, laying the fundamental framework for modern elasticity theory [1].

In the modern language of tensor analysis, the generalized Hooke's law, often simply referred to as Hooke's law, can be written as

$$\sigma^{\alpha\beta} = A^{\alpha\beta\gamma\delta} u_{\gamma\delta} \quad (1.1)$$

where $\sigma^{\alpha\beta}$ is the stress tensor, $u_{\alpha\beta}$ is the strain tensor, and $A^{\alpha\beta\gamma\delta}$ is the elastic tensor. The stress tensor describes internal forces acting across material surfaces, while the strain tensor measures local geometric deformation induced by the displacement field. In linear elasticity theory, the strain tensor can be expressed as $u_{\alpha\beta} = d_{(\alpha,\beta)}$, with the notations $T_{(\alpha\beta)} = \frac{1}{2}(T_{\alpha\beta} + T_{\beta\alpha})$ and $T_{\alpha,\beta} = \partial_\beta T_\alpha$, where d_α is the displacement field. The first notation denotes symmetrization of indices, and the second denotes partial differentiation. This expression also emphasizes that infinitesimal rigid translations and rotations do not cost elastic energy; only the symmetric part



of the displacement gradient contributes to strain.

In a homogeneous and isotropic material, symmetry strongly restricts the form of the elastic tensor. It can be expressed in terms of two Lamé coefficients as

$$A^{\alpha\beta\gamma\delta} = \lambda g^{\alpha\beta} g^{\gamma\delta} + \mu (g^{\alpha\gamma} g^{\beta\delta} + g^{\alpha\delta} g^{\beta\gamma}) \quad (1.2)$$

where g is the metric tensor, $g^{\alpha\beta} = \delta^{\alpha\beta}$ in Euclidean space, and λ and μ are the Lamé coefficients. Roughly speaking, these two coefficients characterize the energetic cost of volumetric and shear deformations, respectively. In the two-dimensional case, they can be rewritten in terms of the experimentally familiar Young's modulus Y and Poisson's ratio ν :

$$\lambda = \frac{\nu Y}{1 - \nu^2}, \quad \mu = \frac{Y}{2(1 + \nu)}. \quad (1.3)$$

At present, elasticity theory has become an important theoretical tool for solving engineering problems and for investigating the physical properties of materials. It also provides the reference point from which the anomalous mechanical response of disordered and amorphous systems can be understood.

1.2 From crystal to amorphous solid

Amorphous solids are a special class of solid materials lacking long-range atomic order. Unlike crystals, whose atoms are arranged periodically, amorphous solids usually possess only short-range structural order and do not have a lattice that can be continued throughout the whole sample. Because of this structural disorder, their mechanical and thermodynamic behaviors differ markedly from those of crystalline materials [2]. In crystals, many mechanical properties can be described in terms of lattice symmetry, phonon modes, and well-defined lattice defects. In amorphous solids, by contrast, the absence of a regular lattice makes the traditional defect language much less direct. Developing an appropriate theoretical description of amorphous mechanics is therefore an important problem in soft matter physics and the physics of disordered systems.

In recent years, plastic events and shear bands in amorphous solids have become major topics of research. A plastic event refers to a local irreversible rearrangement triggered by



an external perturbation. Such local events may accumulate and interact spatially, eventually producing macroscopic shear bands and influencing yielding or mechanical failure. The investigation and characterization of key physical parameters, including the elastic modulus and Poisson's ratio, are therefore of great significance for the development of amorphous materials such as glasses, granular materials, and colloids. These parameters describe the small-strain response, but they also provide a starting point for understanding more complicated plastic and yielding behaviors.

1.2.1 Topological charges and Plasticity

Different defects, or topological charges, can be defined in crystals, such as the dislocations and disclinations shown in Fig. 1.1. These defects represent lattice mismatches that cannot be removed by smooth elastic deformation. With these two types of defects, many behaviors in crystals, including melting, can be well explained and connected to fracton tensor field theory [3]. They play a fundamental role in crystalline systems, including two-dimensional melting and plasticity. The former process is analogous in spirit to the anomalous elasticity of amorphous solids, which will be introduced in the final section of this chapter and in Chapter 2. Moreover, crystal defects carry clear geometric interpretations. For example, dislocations are associated with translational incompatibility, while disclinations are related to angular or curvature defects. This geometric viewpoint motivates the development of the theory of geometric charges in amorphous solids [4][5]. Disclinations and dislocations can be identified by Frank vectors and Burgers vectors, respectively. These vectors give quantitative measures of the strength and orientation of the corresponding defects. However, in contrast to crystals, amorphous solids do not possess a regular atomic lattice. Consequently, distinct defects in the traditional crystalline sense, such as dislocations and disclinations, neither appear clearly during deformation nor can be directly recognized in the initial structure. This does not mean that defect-like mechanical excitations are absent. Rather, it means that new field variables and geometric diagnostics are needed to characterize plastic activity in amorphous materials. It is worth mentioning that a recent method identifies plastic events in amorphous solids through topological charges defined by generalized Burgers vectors in either the displacement field or the eigenvector field [7][8].

While the linear theory of elasticity is undoubtedly applicable to perfectly elastic materials

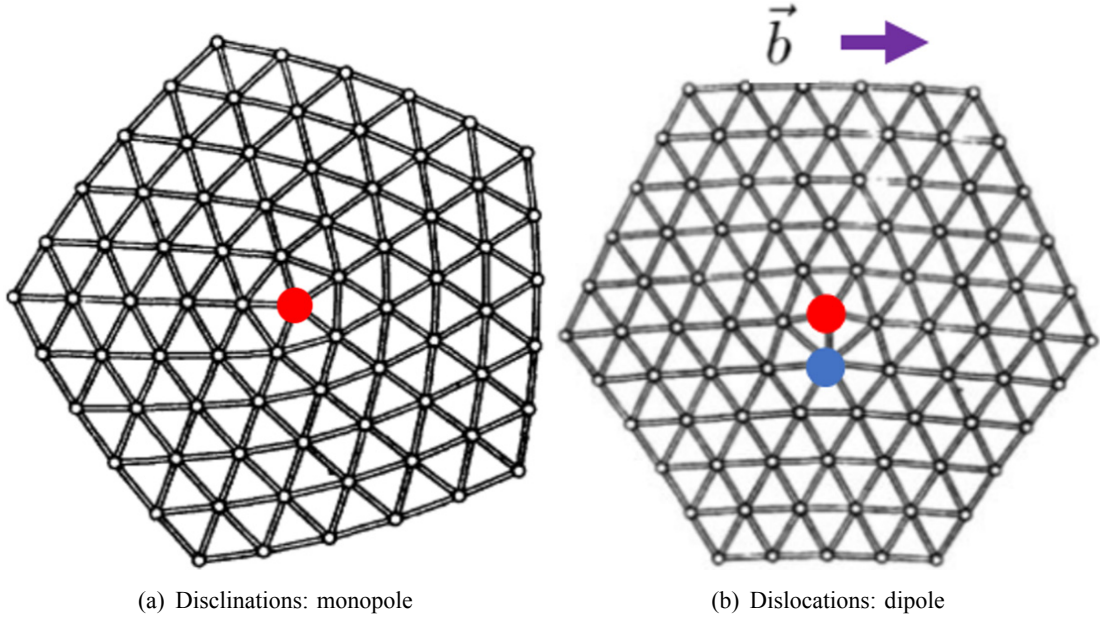


FIG. 1.1 Topological charges in crystals. (Pictures of lattice defects adapted from [6])

under small-strain conditions, its validity for amorphous solids remains debated. For a crystal or an ideal elastic continuum, sufficiently small deformation can often be treated by linear elastic equations. In amorphous solids, however, structural disorder makes local irreversible rearrangements much easier to activate. Decades of research have shown that quasi-localized plastic events occur under any infinitesimal deformation in the thermodynamic limit, leading to nonlinear elastic effects [9]. Thus, even when the macroscopic strain appears small, local plastic activity may still arise and modify the effective mechanical response. The accumulation of such plastic events ultimately forms shear bands under external shear loading. The Eshelby inclusion model, shown in Fig. 1.2, provides a theoretical solution for simple disordered settings and reveals that plastic events can be regarded as quadrupolar singularities in the displacement field [10][11]. To address nonlinear elasticity in amorphous solids more comprehensively, elasto-plastic models have been proposed. These models provide a coarse-grained mesoscopic description of the deformation behavior of amorphous solids [14]. In the athermal quasi-static (AQS) limit of deformation protocols, elasto-plastic models can be viewed as parameter-governed cellular automata (CA), where the relevant parameters include scalar stress and a phenomenological indicator of plastic activity [13]. Such models describe the mechanical response of amorphous systems reasonably well under suitable conditions. Nevertheless,

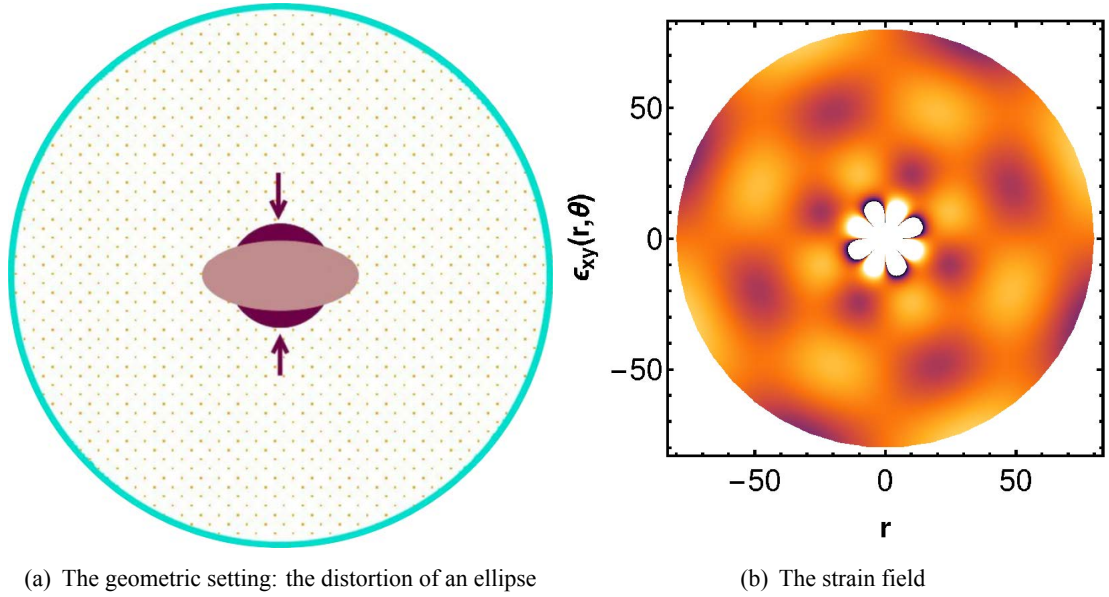


FIG. 1.2 The Eshelby problem in amorphous solids. (Picture adapted from [12] and [13])

several open questions associated with quadrupolar defects remain unresolved, including the Boson peak in glassy systems caused by disorder. In recent years, a new theoretical framework has been proposed in which the free energy is minimized with respect to both the displacement field and quadrupole density in the continuum limit. This framework offers a useful perspective on these issues, namely the field theory introduced below [15].

1.2.2 Jamming point

For athermal systems, amorphous solids, especially granular materials, exhibit distinct mechanical behavior in the unjammed and jammed phases. The jamming point separates these two phases. Near the jamming point, or jamming transition, athermal particles transform from a flowable state into a mechanically stable state. This transition can be identified by a particular packing fraction ϕ_J , called the jamming density [16]. At low density, the system lacks a sufficient contact network to sustain external loads and behaves like an unjammed fluid-like material. As the density increases and approaches the jamming point, a contact network forms and provides mechanical stability.

There are three relevant regimes: the quasilastic phase, the plastic solid phase, and the unjammed phase. They can be distinguished by the pressure p and the packing fraction ϕ . The parameters satisfy $p > p_c$ and $\phi > \phi_c$ in the quasilastic phase, $0 < p < p_c$ and $\phi_J < \phi < \phi_c$



in the plastic solid phase, and $p = 0$ and $\phi < \phi_J$ in the unjammed phase. Correspondingly, two transitions are involved: the jamming transition and the plastic-to-elastic (PE) transition [17]. The density of quadrupoles increases with packing fraction or pressure [5][17]. Unlike thermal phase transitions governed mainly by temperature variations, the jamming transition in granular systems is dominated primarily by geometric constraints. The spatial arrangement of particles, their contact number, and the available volume together determine whether a stable mechanical network can be formed.

In studies of the jamming transition, a key signature of the jamming point is the excess coordination number $\Delta Z \equiv Z - Z^*$, where the coordination number is $Z^* = 2d$ in d -dimensional systems ($Z^* = 4$ for two-dimensional granular disks) [18]. At the jamming point $\phi = \phi_J$, the average coordination number Z equals Z^* . This means that the system has just enough contact constraints to maintain mechanical stability. Under certain conditions, two universal scaling laws appear near the jamming point:

$$p \sim (\phi - \phi_J)^{3/2} \sim \Delta Z^3, \quad \xi \sim (\phi_J - \phi)^{-\nu}, \quad (1.4)$$

where ξ is the correlation length [19][17]. These scaling laws connect pressure, coordination, and spatial correlations close to jamming. As ϕ increases, the dominant multipoles in amorphous solids transform from monopoles to dipoles and quadrupoles. This physical picture will be developed throughout the thesis, with particular emphasis on monopole screening in the pre-jamming phase.

1.3 Theory of geometric charges in amorphous solids

In recent years, Procaccia and co-workers have proposed the concepts of anomalous elasticity and plastic screening to characterize the mechanical properties of disordered solids, in particular granular packings [15][20][21][22] [12][23][24]. They developed a method for describing anomalous elasticity based on displacement fields and geometric defects, including quadrupoles, dipoles, and monopoles. Formally, this approach is analogous to the well-established Debye-Hückel theory for electrolytes and plasmas [25]. In electrostatics, collective charge responses screen long-range Coulomb interactions. In amorphous solids, collective plastic rearrangements may similarly screen long-range elastic response.



Despite the remarkable progress in understanding quadrupolar plastic events and dipole screening effects in amorphous solids, the mechanical response of the unjammed or pre-jamming regimes remains much less understood. In particular, anomalous elasticity in the quasielastic and plastic solid phases can already be described successfully by quadrupole renormalization and dipole screening theories. By contrast, no clear continuum description has yet been established for highly disordered systems below the jamming transition. This issue becomes especially important near the jamming point, where long-range elastic rigidity gradually disappears while strong spatial fluctuations emerge. From the physical perspective, the pre-jamming regime is expected to exhibit screening behavior characterized by a new length scale, rather than the long-range elastic response observed in ordinary solids. Such behavior naturally suggests an effective screening mechanism stronger than dipole screening. Motivated by these observations, recent geometric approaches to amorphous elasticity have proposed that effective monopole-like excitations may emerge from high densities of overlapping quadrupolar plastic events. In this picture, monopole screening provides a possible continuum description for the crossover between unjammed fluid-like states and jammed elastic solids. The main purpose of this thesis is therefore to investigate the physical consequences of monopole screening effects and their relation to the jamming transition.

To better introduce the theory of geometric charges in amorphous solids, we first recall electrostatic theory.

1.3.1 Electrostatic theory

In two-dimensional electrostatic systems, the free energy is given by

$$F = \int dV(\mathcal{U} - \mathcal{W}), \quad \mathbf{E} = -\nabla\phi, \quad (1.5)$$

where $\mathbf{E}$ is the electric field and ϕ is the electric potential. The potential energy density $\mathcal{U}$ and the work density $\mathcal{W}$ in the system, expressed in terms of the electric field $\mathbf{E}$, electric potential ϕ , free charge density ρ_f , and polarization $\mathbf{P}$, are written as

$$\mathcal{U} = \frac{1}{2}\epsilon_0\mathbf{E}^2 + \mathbf{E} \cdot \mathbf{P}, \quad \mathcal{W} = \rho_f\phi + \frac{1}{2}\alpha_2\mathbf{P}^2 + \frac{1}{2}\beta_2(\nabla \cdot \mathbf{P})^2 + \cdots, \quad (1.6)$$



with $\rho = \rho_f + \nabla \cdot \mathbf{P} + \nabla \nabla Q + \dots$. Here, ρ denotes the total charge density, Q is the quadrupole density, $\epsilon = \epsilon_0(1 + \chi_e)$, $\alpha_2 = 1/(\epsilon_0\chi_e)$, and $\beta_2 = 1/l_0^2$. This expansion reflects the fact that a charge distribution may be organized in a multipolar hierarchy.

We define an intrinsic length scale $l \equiv \sqrt{\beta_2/\alpha_2}$ and the system size L . For $l \ll L$, the equilibrium equation derived by minimizing the free energy Eq. (1.5) is the Poisson equation $\Delta\phi = -\rho_f/\epsilon$. This is the usual long-range potential response produced by free charges. In contrast, for $l \gg L$, the equilibrium equation becomes the Helmholtz equation $\Delta\phi - l_0^{-2}\phi = -\rho_f/\epsilon_0$ from Debye-Hückel theory. This equation describes monopole screening in an ionic liquid. The potential is no longer purely long-ranged; instead it decays over a characteristic screening length. This electrostatic screening mechanism is formally analogous to the dipole and monopole screening effects in amorphous solids introduced below.

1.3.2 Free energy, displacement fields and quadrupoles

In this section and the subsequent parts of this thesis, we discuss and simulate amorphous granular solids within an annulus of radii r_{in} and r_{out} . This annular geometry allows a local radial perturbation to be imposed at the inner boundary while the outer boundary is fixed. The resulting displacement profile then reveals how the perturbation propagates and decays through the system. The boundary conditions are specified as $d_r(r_{\text{in}}) = d_0$ and $d_r(r_{\text{out}}) = 0$ for any θ , where the displacement field is $\mathbf{d}(r, \theta) = d_r(r, \theta)\hat{\mathbf{r}} + d_\theta(r, \theta)\hat{\boldsymbol{\theta}}$ (see Fig. 1.3).

In a system Ω , the total free energy F can be divided into the bulk energy density $\mathcal{L}$ and the boundary work density $\mathcal{W}_B$:

$$F = \int_{\Omega} d^2x \mathcal{L} - \oint_{\partial\Omega} dx \mathcal{W}_B, \quad \mathcal{L} = \mathcal{U} - \mathcal{W}, \quad \mathcal{U} = \mathcal{U}_{el} + \mathcal{U}_{Q-el} + \mathcal{U}_{QQ}, \quad \mathcal{W}_B = t^\mu d_\mu, \quad (1.7)$$

where t^μ is the traction force and $\mathcal{W} = 0$ denotes the absence of bulk work. The internal energy density $\mathcal{U}$ can be expressed term by term as

$$\mathcal{U} = \frac{1}{2} A^{\alpha\beta\gamma\delta} u_{\alpha\beta} u_{\gamma\delta} + \mathcal{F}_1(Q^{\alpha\beta}, P^\alpha, \dots) + \mathcal{F}_2(Q^{\alpha\beta}, P^\alpha, \dots), \quad (1.8)$$

where $Q^{\alpha\beta}$ is the introduced quadrupole density describing Eshelby-like deformation, and $P^\alpha = \partial_\beta Q^{\alpha\beta}$ is defined as the dipole density. The first term is the ordinary elastic energy, while the

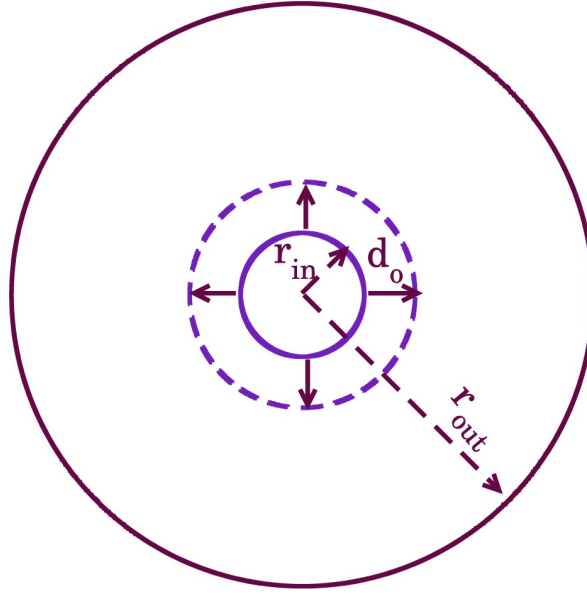


FIG. 1.3 The model used in [15] and in this thesis. (Picture adapted from [26])

remaining terms describe couplings between geometric charges and the elastic field, as well as the energetic cost of the geometric charges themselves. By setting $\delta_d F = \delta_Q F = 0$ to minimize the free energy Eq. (1.7), one obtains the equilibrium equations of the system. For example, in purely elastic systems we have $\sigma^{\alpha\beta} n_\beta|_{\partial\Omega} = t^\alpha$, $\partial_\alpha \sigma^{\alpha\beta} = 0$, and, under polar symmetry, $d_r(r) = d_0 \frac{r^2 - r_{out}^2}{r_{in}^2 - r_{out}^2} \frac{r_{in}}{r}$. This expression gives the classical elastic radial displacement profile in the absence of plastic screening.

In a quasilastic phase with dilute quadrupoles, the dominant terms are taken as $\mathcal{F}_1(Q^{\alpha\beta}) = \Gamma^{\alpha\beta}_{\gamma\delta} u_{\alpha\beta} Q^{\gamma\delta}$ and $\mathcal{F}_2(Q^{\alpha\beta}) = \frac{1}{2} \Lambda_{\alpha\beta\gamma\delta} Q^{\alpha\beta} Q^{\gamma\delta}$. The equilibrium equations read

$$Q^{\alpha\beta} = -\tilde{\Lambda}^{\alpha\beta\gamma\delta} u_{\gamma\delta}, \quad \partial_\alpha \tilde{\sigma}^{\alpha\beta} = 0 \quad (1.9)$$

where $\tilde{\Lambda}^{\alpha\beta\gamma\delta} \equiv -\Lambda^{\alpha\beta\mu\nu} \Gamma^{\gamma\delta}_{\mu\nu} u_{\gamma\delta}$ and $\tilde{\sigma}^{\alpha\beta} = \sigma^{\alpha\beta} + \Gamma^{\alpha\beta}_{\gamma\delta} Q^{\gamma\delta}$. Here, $\Lambda^{\alpha\beta\mu\nu}$ is the inverse of $\Lambda_{\alpha\beta\gamma\delta}$. The quadrupole density is induced by the strain field and, in turn, modifies the effective stress. This is equivalent to a renormalization of the elastic tensor derived from Eq. (1.9):

$$\tilde{A}^{\mu\nu\rho\sigma} = A^{\mu\nu\rho\sigma} + \Lambda_{\alpha\beta\gamma\delta} \tilde{\Lambda}^{\alpha\beta\mu\nu} \tilde{\Lambda}^{\gamma\delta\rho\sigma} - 2\Gamma^{\mu\nu}_{\gamma\delta} \tilde{\Lambda}^{\gamma\delta\rho\sigma} \quad (1.10)$$

In the plastic solid phase or screening regime, where quadrupoles occur at higher densities,



the quadrupole renormalization contribution can be absorbed into effective elastic constants because it does not change the form of the equilibrium equations. The more important effect is then the dipole screening induced by spatial gradients of the quadrupole density. Thus we have $\mathcal{F}_1(P^\alpha) = \Gamma^\alpha_\beta d_\alpha P^\beta$ and $\mathcal{F}_2(P^\alpha) = \frac{1}{2} \Lambda_{\alpha\beta} P^\alpha P^\beta$. The equilibrium equations read

$$P^\alpha = -\Lambda^{\alpha\beta} \Gamma^\gamma_\beta d_\gamma, \quad \partial_\alpha \sigma^{\alpha\beta} = \Gamma^\beta_\gamma P^\gamma \quad (1.11)$$

where $\Lambda^{\alpha\beta}$ is the inverse of $\Lambda_{\alpha\beta}$, $\Gamma^\alpha_\beta = \mu_1 \delta^\alpha_\beta$, and $\Lambda_{\alpha\beta} = \mu_2 \delta_{\alpha\beta}$. Under polar symmetry, these equations can be rewritten as $\Delta \mathbf{d} = -\kappa_P^2 \mathbf{d}$, where $\kappa_P^2 \equiv \frac{\mu_1^2}{\mu_2(\lambda+2\mu)}$. The solution is composed of the Bessel functions J_1 and Y_1 :

$$d_r(r) = d_0 \frac{Y_1(\kappa_P r) J_1(\kappa_P r_{out}) - J_1(\kappa_P r) Y_1(\kappa_P r_{out})}{Y_1(\kappa_P r_{in}) J_1(\kappa_P r_{out}) - J_1(\kappa_P r_{in}) Y_1(\kappa_P r_{out})}, \quad (1.12)$$

which differs from the classical elastic solution and is therefore referred to as anomalous elasticity [21]. The asymptotic behavior of Eq. (1.12) at large r in the disk setting reads

$$d_r(r) \sim \sqrt{\frac{2\ell_P}{\pi r}} \cos\left(\frac{r}{\ell_P} - \frac{n\pi}{2} - \frac{\pi}{4}\right) \sim \frac{1}{\sqrt{r}} \quad (1.13)$$

Thus, dipole screening introduces oscillatory structure, but its envelope remains a relatively weak algebraic decay.

Prior to this thesis, there was no well-established explanation for the physical characteristics of the unjammed phase. In particular, the way in which long-range elastic response is screened below jamming, and how this screening is related to the jamming transition, still required clarification.

1.4 Structure of the thesis

The thesis is organized as follows. In Chapter 2, we reintroduce the geometric theory of anomalous elasticity in amorphous solids and derive the hierarchy of quadrupole, dipole, and monopole screening equations. In Chapter 3, we solve the monopole screening equation in terms of the displacement field and analyze the mathematical structure and asymptotic behavior of the solution. In Chapter 4, we analyze the physical characteristics of monopole screening effects



in the pre-jammed phase based on numerical data obtained using LAMMPS. Finally, we give a brief conclusion.

2 Geometric theory of anomalous elasticity

Moshe and co-workers proposed an extended theoretical framework based on screening effects and differential geometry, attempting to explain the unique behaviors of geometric charges via the Airy potential function. This framework corresponds closely to geometric descriptions of defects in crystalline systems [27][5][4]. In crystals, defects can be interpreted through the incompatibility of the underlying lattice geometry. In amorphous solids, although no regular lattice is available, plastic rearrangements and the stresses induced by them can still be described by continuous geometric fields. Although quadrupoles and dipoles have been investigated extensively for a long time, the physical properties of monopoles remain largely unexplored. Here for completeness, we provide a brief introduction to the geometric theory of amorphous solids, which sheds light on topological charges, particularly monopoles. This theory supplies a unified continuum language for discussing plastic events and their effective multipole descriptions. We then derive the screening equations in terms of the displacement field in planar geometry, where the monopole screening equation serves as the starting point for our numerical simulations.

2.1 Geometric description of defects embedded in a continuum elastic medium

To describe the properties of amorphous solids in a continuum limit, we first assume a manifold $\mathcal{M}$ with a reference metric $\bar{g}^0$ to represent the amorphous solid under no external loads and with rest plastic events (the background field), while an actual metric g describes the system under external loads. The reference metric and the actual metric therefore encode the unloaded and loaded geometric relations of the same material configuration. We then define an embedding $\phi : \mathcal{M} \rightarrow \mathbb{R}^2$ from the actual space of the amorphous solid to the real physical space, where the metric is $\eta_{\mu\nu} = \delta_{\mu\nu}$ in Euclidean space:

$$dl_0^2 = \bar{g}_{\mu\nu}^0 dx^\mu dx^\nu, \quad dl^2 = g_{\mu\nu} dx^\mu dx^\nu, \quad dl^2 = \eta_{\alpha\beta} d\tilde{x}^\alpha d\tilde{x}^\beta = \eta_{\alpha\beta} \partial_\mu \phi^\alpha \partial_\nu \phi^\beta dx^\mu dx^\nu. \quad (2.1)$$

Thus, the metric g can be expressed as $g = \nabla \phi^T \nabla \phi$. This relation shows that the metric records how the embedding changes infinitesimal lengths and angles on the manifold. To construct the elastic theory on $\mathcal{M}$, the displacement field is defined as $d_\mu = g_{\mu\nu}(\phi^\nu - \phi_*^\nu)$, and the strain tensor is



defined as $u_{\mu\nu} = \frac{1}{2}(g_{\mu\nu} - \bar{g}_{\mu\nu}^0)$. Here ϕ_* denotes the stress-free, energy-minimizing configuration. In this form, elastic deformation is tied directly to the difference between the actual metric and the reference metric. The elastic part u^{el} and the plastic part u^{pl} of the strain tensor are defined respectively as $u_{\mu\nu}^{\text{el}} = \frac{1}{2}(g_{\mu\nu} - \bar{g}_{\mu\nu})$ and $u_{\mu\nu}^{\text{pl}} = \frac{1}{2}q_{\mu\nu}$ where q represents the quadrupole contribution in the metric:

$$\bar{g}_{\mu\nu} = \bar{g}_{\mu\nu}^0 + q_{\mu\nu}, \quad (2.2)$$

which can be understood as the origin of quadrupoles being the basic unit of plastic events from the geometric perspective. Equivalently, a plastic event is represented not merely as a displacement in the external coordinates, but as a change in the natural internal geometry of the material. For pure elasticity, the quadrupole contribution vanishes. In contrast, in a plastic solid phase or an unjammed phase, the quadrupole density can be defined as $Q^{\alpha\beta} = \bar{\epsilon}^{\alpha\mu} \bar{\epsilon}^{\beta\nu} q_{\mu\nu}$ where $\bar{\epsilon}$ is the Levi-Civita symbol on the manifold corresponding to the metric $\bar{g}^0$. This tensor density reorganizes metric perturbations into quantities that have the meaning of geometric charges.

From the metric, we can derive the Gaussian curvature K and the reference Gaussian curvature K^0 , both of which can be expressed in terms of multipoles. Here, the geometric compatibility condition $K = 0$ is given by the Cauchy-Riemann condition of the strain $u_{\alpha\beta}$. The reference Gaussian curvature K^0 can be interpreted as the curvature of the real space in the experimental setup. In this way, plastic defects and elastic response are brought into a single geometric description through the relation between metric and curvature.

2.2 Elasticity

We construct the free energy to describe the elasticity in two-dimensional amorphous solids as

$$F = \int_{\mathcal{M}} dS_{\bar{g}^0} \mathcal{L} - \int_{\partial\mathcal{M}} dl_{\bar{g}^0} \mathcal{W}_B, \quad \mathcal{L} = \mathcal{U} - \mathcal{W}, \quad (2.3)$$

where $\mathcal{U}$ denotes the elastic energy density, $\mathcal{W}$ the bulk work density, and $\mathcal{W}_B$ the boundary work density. The integration measure is determined by the reference metric $\bar{g}^0$, so the free energy is defined on the material reference configuration. The elastic energy density is just



Hookean and $\mathcal{W}_B$ arises from the traction force acting on the boundary

$$\mathcal{U} = \frac{1}{2} A^{\alpha\beta\gamma\delta} u_{\alpha\beta}^{\text{el}} u_{\gamma\delta}^{\text{el}}, \quad \mathcal{W}_B = \mathbf{t} \cdot \mathbf{d}. \quad (2.4)$$

with the elastic stress given by $\sigma_{\text{el}}^{\alpha\beta} = A^{\alpha\beta\gamma\delta} u_{\gamma\delta}^{\text{el}}$. This definition is consistent with the usual linear stress-strain relation, except that the strain is expressed through metric differences.

In a purely elastic system, the equilibrium equations are derived by minimizing the free energy Eq. (2.3) as $\delta_\phi F = 0$ with $\mathcal{W} = 0$ and $q = 0$. The resulting equilibrium equations read

$$\bar{\nabla}_\mu \sigma_{\text{el}}^{\mu\nu} + (\Gamma_{\alpha\beta}^\nu - \bar{\Gamma}_{\alpha\beta}^\nu) \sigma_{\text{el}}^{\alpha\beta} = 0, \quad n_\alpha \sigma_{\text{el}}^{\alpha\beta} = t^\beta, \quad (2.5)$$

where Γ and $\bar{\Gamma}$ denote the Christoffel symbols corresponding to the metric g and $\bar{g}^0$, respectively, and $\bar{\nabla}$ is the covariant derivative compatible with respect to the metric $\bar{g}^0$. The first equation represents mechanical equilibrium in the bulk, while the second one imposes the traction condition on the boundary. Compared with the ordinary equilibrium equation in flat space, the additional Christoffel-symbol term records the geometric mismatch between the actual and reference configurations.

2.3 Equilibrium Equations

For systems containing plastic events, the quadrupole contribution q is nonzero, and the bulk work density $\mathcal{W}[q]$ is expressed via a multipole expansion of quadrupoles, dipoles and monopoles:

$$\mathcal{W} = \frac{1}{2} \Lambda_{\alpha\beta\gamma\delta}^Q Q^{\alpha\beta} Q^{\gamma\delta} + \frac{1}{2} \Lambda_{\alpha\beta}^P P^\alpha P^\beta + \frac{1}{2} \Lambda^M M^2, \quad (2.6)$$

where the dipole density is defined as $P^\alpha = \bar{\nabla}_\beta Q^{\alpha\beta}$ and the monopole density is defined as $M = \bar{\nabla}_{\alpha\beta} Q^{\alpha\beta}$. Note that dipoles and monopoles are derived from quadrupoles. As the density of quadrupole increases, effective dipoles and subsequently monopoles are excited, since higher-order derivatives of the quadrupole density become non-negligible at that time. By minimizing the free energy with respect to the embedding ϕ and quadrupole q , we obtain the equilibrium equations.

From the physical viewpoint, the hierarchy of quadrupoles, dipoles and monopoles can be interpreted as different effective descriptions of plastic activity at different disorder strengths



and packing fractions. Quadrupoles correspond to localized Eshelby-like shear transformations, which dominate in quasielastic systems where plastic events remain dilute and weakly interacting. In this regime, the displacement field still preserves a relatively long-range elastic character. As the density of plastic events increases, spatial gradients of the quadrupole density become non-negligible. Effective dipole excitations are then generated through the relation between densities of dipoles and quadrupoles. Dipole screening partially suppresses long-range elastic propagation and introduces oscillatory screening behavior in the displacement field, corresponding to the plastic-solid phase discussed previously. When the system enters the pre-jamming regime, plastic events become highly collective and strongly overlap spatially. In this situation, second-order gradients of the quadrupole density become increasingly important, leading to the emergence of effective monopole excitations. Unlike quadrupoles and dipoles, monopoles generate much stronger screening effects and naturally introduce screening lengths into the mechanical response. Consequently, displacement fields exhibit exponentially damped propagation instead of purely algebraic elastic decay. This distinction constitutes the essential physical difference between ordinary elasticity and monopole-dominated anomalous elasticity near the jamming transition.

Furthermore, in homogeneous and isotropic systems, the coupling constants read

$$\Lambda_{\alpha\beta\gamma\delta}^Q = \lambda_Q \bar{g}_{\alpha\beta}^0 \bar{g}_{\gamma\delta}^0 + \mu_Q (\bar{g}_{\alpha\gamma}^0 \bar{g}_{\beta\delta}^0 + \bar{g}_{\alpha\delta}^0 \bar{g}_{\beta\gamma}^0), \quad (2.7)$$

$$\Lambda_{\alpha\beta}^P = \frac{1}{2} Y \ell_P^2 \bar{g}_{\alpha\beta}^0, \quad (2.8)$$

$$\Lambda^M = Y \ell_M^4, \quad (2.9)$$

where ℓ_P and ℓ_M are the typical length scales associated with each screening multipole. The wave numbers κ_P and κ_M are the coupling constants defined as the inverses of the typical length scales. These length scales determine how far the corresponding screening contribution remains effective in space. The effects of quadrupole renormalization, dipole screening and monopole screening dominate the system behavior respectively as the density of quadrupoles increases from small to large. These effects correspond respectively to quasielastic phase, the plastic solid phase and the pre-jamming phase (the unjammed phase near the jamming point, which will be elaborated in the following chapter).



2.3.1 Quadrupole

In the quasielastic phase governed with quadrupoles, the work density Eq. (2.6) is dominated by the quadrupole contribution

$$\mathcal{W} = \frac{1}{2} \Lambda_{\alpha\beta\gamma\delta}^Q Q^{\alpha\beta} Q^{\gamma\delta}. \quad (2.10)$$

The corresponding equilibrium equation yields a linear relation between the elastic stress and the quadrupole density as

$$\sigma_{\text{el}}^{\alpha\beta} + 2\bar{\varepsilon}^{\alpha\mu}\bar{\varepsilon}^{\beta\nu}\Lambda_{\mu\nu\gamma\delta}^Q Q^{\gamma\delta} = 0, \quad (2.11)$$

which, when compared with the equilibrium form for purely elastic systems in Eq. (2.5), describes the renormalization effect of the elastic moduli in Eq. (1.2). In this regime, quadrupoles do not introduce a new differential structure into the equilibrium equations; instead, their effect is absorbed into effective elastic constants. This gives the renormalized elastic tensor

$$\tilde{A}^{\alpha\beta\gamma\delta} = \Gamma^{-1}_{\mu\nu}{}^{\alpha\beta} A^{\mu\nu\gamma\delta}, \quad \Gamma^{\alpha\beta}{}_{\mu\nu} = I^{\alpha\beta}{}_{\mu\nu} - \frac{1}{4} A^{\alpha\beta\gamma\delta} \Lambda_{\gamma\delta\mu\nu}^q, \quad (2.12)$$

where $I^{\alpha\beta}{}_{\mu\nu} = \frac{1}{2}(\delta^{\alpha}_{\mu}\delta^{\beta}_{\nu} + \delta^{\alpha}_{\nu}\delta^{\beta}_{\mu})$ denotes the (2,2)-rank identity operator, and $\Lambda_{\alpha\beta\gamma\delta}^q$ is the inverse of $\Lambda_q^{\alpha\beta\gamma\delta} = \Lambda_{\mu\nu\rho\sigma}^Q \bar{\varepsilon}^{\mu\alpha}\bar{\varepsilon}^{\nu\beta}\bar{\varepsilon}^{\rho\gamma}\bar{\varepsilon}^{\sigma\delta}$, which normalizes the elastic stress tensor.

Since quadrupole screening does not alter the form of the equilibrium equations, we replace the effective elastic modulus $\tilde{A}$ with the original notation A for convenience in subsequent discussions. This convention allows us to focus on the new screening structures introduced by dipoles and monopoles.

2.3.2 Dipole

In the plastic solid phase featuring dipole screening effects, the work density Eq. (2.6) is dominated by the dipole contribution

$$\mathcal{W} = \frac{1}{2} \Lambda_{\alpha\beta}^P P^{\alpha} P^{\beta}. \quad (2.13)$$



Since the dipole density is the first spatial derivative of the quadrupole density, this contribution measures the energetic cost of spatially varying plastic activity. The corresponding equilibrium equations read

$$\sigma_{\text{el}}^{\alpha\beta} + Y\ell_P^2 \bar{\epsilon}^{\mu\alpha} \bar{\epsilon}^{\nu\beta} \bar{\nabla}_{(\mu} P_{\nu)} = 0, \quad \bar{\epsilon}^{\mu\alpha} \bar{\epsilon}^{\nu\beta} n_{(\mu} P_{\nu)} = 0. \quad (2.14)$$

where $P_\mu = \bar{g}_{\mu\nu}^0 P^\nu$, and ℓ_P is given in Eq. (2.8). The first equation balances elastic stress against gradients of the dipole density, and the second equation gives the corresponding compatibility condition at the boundary.

2.3.3 Monopole

In the pre-jamming phase (the unjammed phase near the jamming point) with monopole screening effect, the work density Eq. (2.6) is dominated by the monopole contribution

$$\mathcal{W} = \frac{1}{2} \Lambda^M M^2. \quad (2.15)$$

The monopole density is the second spatial derivative of the quadrupole density and therefore captures a higher-order spatial inhomogeneity of plastic events. The corresponding equilibrium equations are

$$\sigma_{\text{el}}^{\alpha\beta} + Y\ell_M^4 \bar{\epsilon}^{\mu\alpha} \bar{\epsilon}^{\nu\beta} \bar{\nabla}_{\mu\nu} M = 0, \quad \bar{\epsilon}^{\mu\alpha} \bar{\epsilon}^{\nu\beta} n_{(\mu} \bar{\nabla}_{\nu)} M = 0. \quad (2.16)$$

where $M = \bar{\nabla}_{\alpha\beta} Q^{\alpha\beta}$ is the monopole density and ℓ_M is given in Eq. (2.9). Compared with the dipole equation, the monopole equation contains a higher-order derivative structure, which is why it produces a stronger screening response.

2.4 Screening equations

To derive the geometric screening equations from the equilibrium equations, we need to construct basic equations of the Gaussian curvature K and the metric g in the actual space. From Eq. (2.2) and the definition of stress tensor, the actual metric can be expressed as

$$g_{\alpha\beta} = \bar{g}_{\alpha\beta}^0 + \bar{\epsilon}_{\alpha\mu} \bar{\epsilon}_{\beta\nu} Q^{\mu\nu} + 2A_{\alpha\beta\gamma\delta} \sigma_{\text{el}}^{\gamma\delta}, \quad (2.17)$$



where $A_{\alpha\beta\gamma\delta}$ denotes the inverse of the elastic modulus tensor $A^{\alpha\beta\gamma\delta}$. This equation states that the actual metric consists of the reference metric, the plastic contribution induced by geometric charges, and the elastic contribution induced by stress. In real-world experiments and numerical simulations, we make linear approximations, namely $g \approx \bar{g} \approx \bar{g}^0$ and perform linearization of the perturbation expansion for the Gaussian curvature K under metric perturbations of g . This approximation turns a generally nonlinear geometric problem into a tractable linear screening theory.

2.4.1 Airy stress function

To solve the equilibrium equations (Eq. (2.11), Eq. (2.14) and Eq. (2.16)), we first introduce the Airy stress function χ within a geometric framework

$$\sigma_{\text{el}}^{\alpha\beta} = \frac{1}{\sqrt{|g|}} \frac{1}{\sqrt{|g|}} \varepsilon^{\alpha\mu} \varepsilon^{\beta\nu} \nabla_{\mu\nu} \chi. \quad (2.18)$$

where ε is the Levi-Civita symbol and ∇ denotes the covariant derivative, both defined with respect to the actual metric g . The use of the Airy function automatically incorporates the force-balance structure and reduces the problem to an equation for a scalar potential.

By computing the Gaussian curvature K associated with the actual metric g under linear approximations and the condition $\bar{K}^0 \approx 0$, we derive the original screening equations with the geometric compatibility condition by substituting Eq. (2.18) into Eq. (2.17):

$$0 = \bar{K}^0 + \bar{\nabla}_{\alpha\beta} Q^{\alpha\beta} - \frac{1}{\bar{Y}} \Delta \chi \quad (2.19)$$

where the induced effective monopole density $\bar{\nabla}_{\alpha\beta} Q^{\alpha\beta}$ is given by

$$\bar{\nabla}_{\alpha\beta} Q^{\alpha\beta} = -\frac{1}{\bar{Y}} \begin{cases} 0, & \text{Quadrupole;} \\ \ell_P^{-2} \bar{\Delta} \chi, & \text{Dipole;} \\ \ell_M^{-4} \chi, & \text{Monopole.} \end{cases} \quad (2.20)$$

by substituting Eq. (2.18) into Eq. (2.11), Eq. (2.14) and Eq. (2.16) with a trivial gauge choice. Here, $\bar{Y}$ is the renormalized Young's modulus, Δ is the Laplacian operator with respect to the



metric g while $\bar{\Delta}$ is with respect to the reference metric $\bar{g}^0$. Three distinct screening equations corresponding to the three types of screening effects are obtained by substituting the induced monopole density Eq. (2.20) into the curvature equation Eq. (2.19):

$$\frac{1}{\bar{Y}}\Delta\Delta\chi = \bar{K}^0, \quad \text{Quadrupole;} \quad (2.21)$$

$$\frac{1}{\bar{Y}}\Delta\Delta\chi + \frac{1}{\bar{Y}}\ell_P^{-2}\Delta\chi = \bar{K}^0, \quad \text{Dipole;} \quad (2.22)$$

$$\frac{1}{\bar{Y}}\Delta\Delta\chi + \frac{1}{\bar{Y}}\ell_M^{-4}\chi = \bar{K}^0, \quad \text{Monopole;} \quad (2.23)$$

which can be combined into a unified screening equation:

$$\frac{1}{\bar{Y}}\Delta\Delta\chi + \frac{1}{\bar{Y}}\ell_P^{-2}\Delta\chi + \frac{1}{\bar{Y}}\ell_M^{-4}\chi = \bar{K}^0, \quad (2.24)$$

where the typical length scales control the contributions of different terms in the screening equations, namely the effective screening effects of the corresponding multipoles in amorphous solids. The unified equation also emphasizes that quadrupole, dipole and monopole descriptions are not independent theories, but successive approximations within the same geometric screening framework.

2.4.2 Displacement fields in planar geometry

In this subsection, we reformulate the screening equations in terms of the displacement field $\mathbf{d}$ for planar geometry where $\bar{K}^0 = 0$. To transform the potential equations to the form of the displacement field, we need to first obtain the relation between the Airy stress function χ and the displacement field $\mathbf{d}$ from Eq. (1.1), Eq. (1.2), and Eq. (2.18):

$$\nabla \cdot \mathbf{d} = \frac{1}{\bar{Y}'}\Delta\chi \quad (2.25)$$

where the effective Young's modulus is defined as $\bar{Y}' = 2\lambda + 2\mu$. Here, in the planar geometry, different differential operators are approximately equivalent ($\nabla \approx \partial$), and tensors defined with respect to g and $\bar{g}^0$ are also approximately identical ($\bar{\nabla} \approx \nabla$ and $\bar{\varepsilon} \approx \varepsilon$). This conversion is important because the numerical simulations directly measure particle displacements rather than the Airy stress function.



Substituting Eq. (2.25) into the screening equations (Eq. (2.21), Eq. (2.22) and Eq. (2.23)), we can derive the screening equations in terms of the displacement field $\mathbf{d}$:

$$\Delta\Delta\mathbf{d} = 0, \quad \text{Quadrupole;} \quad (2.26)$$

$$\Delta\Delta\mathbf{d} + \ell_p^{-2}\Delta\mathbf{d} = 0, \quad \text{Dipole;} \quad (2.27)$$

$$\Delta\Delta\mathbf{d} + \ell_M^{-4}\mathbf{d} = 0, \quad \text{Monopole;} \quad (2.28)$$

where the last equation is with a trivial gauge choice. Here, the screening equations should be understood as a linear theory. It constitutes the central governing equation for subsequent analysis. Detailed derivations for this subsection are provided in Appendix A. The three screening equations above reveal qualitatively different mechanical behaviors in amorphous solids. The quadrupole equation preserves the long-range nature of elasticity and corresponds to ordinary biharmonic elastic propagation. Dipole screening modifies the elastic response by introducing oscillatory behavior with weak screening effects, while the monopole equation fundamentally changes the character of the displacement field through the appearance of a finite screening length. In particular, the monopole screening equation predicts exponentially damped displacement fields at sufficiently large distances. Such exponential decay indicates that mechanical perturbations can no longer propagate throughout the entire system efficiently, reflecting the loss of long-range rigidity in the pre-jamming regime. This behavior is analogous to screening effects in electrostatic systems, where electric fields become exponentially suppressed due to collective charge responses. In the present context, the screening originates from collective plastic rearrangements and geometric incompatibility induced by disorder.

2.5 Discussion of the geometric theory

Although the geometric theory has been proposed to account for the anomalous elasticity in amorphous solids and successfully applied to numerous scenarios within the quasielastic phase and the plastic solid phase [28], the monopole screening effect in the pre-jamming phase is still not understood fully, both in existing literature and in the present thesis. A central issue arises from nonlinear effects appearing from Eq. (2.19) to Eq. (2.28), which originates from the breakdown of the linear approximation with high densities of plastic events (mainly monopoles) in the



pre-jamming phase. Thus the theory we introduce and use here is linear. The difficulty of solving the nonlinear screening equation is analogous to studies on the Einstein field equation with nonlinear gravitational terms [29]. This motivates us to solve the linearized monopole screening equation in this work, while leaving nonlinear effects for future investigations. It is also worth mentioning that the geometric theory we have shown before excludes fluctuations, in contrast to the ensemble theory of thermodynamics and statistical physics. We therefore interpret the geometric theory as valid for individual single-system realizations only. The interactions between different types of topological charges, namely quadrupole-dipole, quadrupole-monopole and dipole-monopole, are not taken into account here, although intra-type interactions of the same type of topological charges have been included in the theory [5]. Such inter-type interactions remain to be included in future work. These limitations do not remove the usefulness of the linear geometric theory as a starting point; rather, they indicate where the description must be extended in order to fully describe the pre-jamming and unjammed regimes.



3 Solutions of the monopole screening equation

In this chapter, we present and solve the monopole screening equation. Then we analyze the properties of the particular solution with boundary conditions compatible with numerical simulations. Note that the monopole screening equation is valid within the regime of effective linearized Gaussian-curvature approximation (i.e., where the density of plastic events is sufficiently low) in principle. Nevertheless, owing to nonlinear effects, we focus exclusively on monopoles in the pre-jamming phase. The purpose of this chapter is therefore not to solve the full nonlinear geometric theory, but to extract the simplest analytical consequence of monopole screening that can be compared directly with simulation data.

3.1 Equation of monopole screening

The starting point is the monopole screening equation Eq. (2.28), which we restated here for clarity:

$$\Delta\Delta\mathbf{d} + \ell_M^{-4}\mathbf{d} = 0. \quad (3.1)$$

where $\mathbf{d}$ is the displacement field. This equation is a bi-Helmholtz equation, and the monopole length scale ℓ_M determines the strength of the screening term. It is a bi-Helmholtz equation with well-known analytical solutions under certain conditions. Firstly, by assuming polar symmetry $\mathbf{d}(r, \theta) = d_r(r)\hat{\mathbf{r}}$, Eq. (3.1) can be reduced to an ordinary differential equation (ODE) with respect to the radial coordinate r :

$$\left(\frac{d^4}{dr^4} + \frac{2}{r} \frac{d^3}{dr^3} - \frac{3}{r^2} \frac{d^2}{dr^2} + \frac{3}{r^3} \frac{d}{dr} - \frac{3}{r^4} + \ell_M^{-4} \right) d_r(r) = 0. \quad (3.2)$$

Notably, vector-valued ODE derivatives differ from their scalar counterparts, leading to the important extra terms. These extra radial terms originate from the angular dependence of the basis vector $\hat{\mathbf{r}}$ in polar coordinates, even though the scalar amplitude $d_r(r)$ itself is independent of θ .



3.2 Solve the equation of monopole screening

Prior to solving the screening equation Eq. (3.2) with radial displacement field $d_r(r)$, we introduce boundary conditions to determine coefficients, as illustrated in Fig. 1.3. However, the setting of simulations, $d_r(r_{\text{in}}) = d_0$ and $d_r(r_{\text{out}}) = 0$, are insufficient for fourth-order differential equations Eq. (3.1) and Eq. (3.2). Therefore, we need to introduce more boundary conditions. Here we choose the physical condition $d_r(r) \rightarrow 0$ for increasing r ($r_{\text{in}}, \ell_M \ll r < r_{\text{out}}$) to eliminate divergent contributions in the solution of Eq. (3.2), which will be validated by numerical simulations in the subsequent chapter. Furthermore, all boundary conditions read

$$d_r(r_{\text{in}}) = d_0, d_r(r_{\text{out}}) = 0; d_r(r_{\text{in}}, \ell_M \ll r < r_{\text{out}}) \rightarrow 0 \text{ for increasing } r \quad (3.3)$$

The first two conditions match the annular inflation protocol: the inner boundary is displaced outward and the outer boundary is fixed. The last condition selects the screened branch of the solution and excludes terms that grow at large radius. Detailed derivations for this section are provided in Appendix B.

3.2.1 General solution

To solve Eq. (3.2), we transform it into an ODE expressed in terms of Bessel functions with complex wavenumbers, which can further be recast using Kelvin functions. This representation is convenient because Kelvin functions are real-valued combinations of Bessel functions with complex arguments. The resulting general solution reads

$$d_r(r) = C_1 \ker_1(\kappa_M r) + C_2 \text{kei}_1(\kappa_M r) + C_3 \text{ber}_1(\kappa_M r) + C_4 \text{bei}_1(\kappa_M r) \quad (3.4)$$

where $\kappa_M = 1/\ell_M$ is the wavenumber for monopole screening, and $\ker_1$, kei_1 , ber_1 and bei_1 denote first-order Kelvin functions. Note that the first two terms $\ker_1, \text{kei}_1$ in the general solution Eq. (3.4) correspond to oscillatory decaying solutions, whereas the last two terms $\text{ber}_1, \text{bei}_1$ diverge in large- r limit. This difference is essential for the physical selection of solutions in a finite annular system, because only the decaying sector is compatible with a screened mechanical response away from the inner perturbation.



3.2.2 Particular solution with boundary conditions

Using the boundary conditions Eq. (3.3), we obtain the particular solution of Eq. (3.2) from the general form Eq. (3.4). The particular solution can be written as

$$d_r(r) = d_0 \frac{\ker_1(\kappa_M r) \text{kei}_1(\kappa_M r_{\text{out}}) - \text{kei}_1(\kappa_M r) \ker_1(\kappa_M r_{\text{out}})}{\ker_1(\kappa_M r_{\text{in}}) \text{kei}_1(\kappa_M r_{\text{out}}) - \text{kei}_1(\kappa_M r_{\text{in}}) \ker_1(\kappa_M r_{\text{out}})}. \quad (3.5)$$

Here the divergence-free condition enforces $C_3 = C_4 = 0$ in Eq. (3.4). This particular solution shares a similar form with that of the dipole screening equation, given by Eq. (1.12). The normalization is fixed by $d_r(r_{\text{in}}) = d_0$, while the numerator is arranged so that the displacement vanishes at $r = r_{\text{out}}$.

The asymptotic behavior of the particular solution Eq. (3.5) for large radial distance in the annular geometry follows directly from the asymptotic properties of Kelvin functions:

$$d_r(r) \sim \frac{1}{\sqrt{r}} e^{-r/L_M} \sin\left(\frac{r_{\text{out}} - r}{L_M}\right) \sim \frac{1}{\sqrt{r}} e^{-r/L_M} \quad (3.6)$$

where $L_M = \sqrt{2}\ell_M$ is defined as screening length. The normalized displacement field $\sqrt{r}d_r$ Eq. (3.6) exhibits exponential decay for monopole screening, much faster than the amplitude-constant behavior of the dipole screening solution Eq. (1.13). Such exponential decay constitutes the key distinction between monopole and dipole screening effects. The oscillatory sine factor controls sign changes and finite-size interference, whereas the exponential factor controls the envelope of the response.

3.3 Properties of the particular solution

Following the standard analysis procedure for dipole-screening solutions, we investigate the properties of the particular solution Eq. (3.5) under appropriate boundary conditions in this section. Theoretical derivations of these properties are provided in Appendix B. The analysis below is useful for interpreting fitting results, because it identifies which features come from the mathematical solution itself and which features should be attributed to simulation details.

By plotting solutions Eq. (3.5) for different wavenumbers κ_M , we visualize its key feature, as shown in Fig. 3.1. The plotted quantity rd_r makes the oscillatory part more visible, while

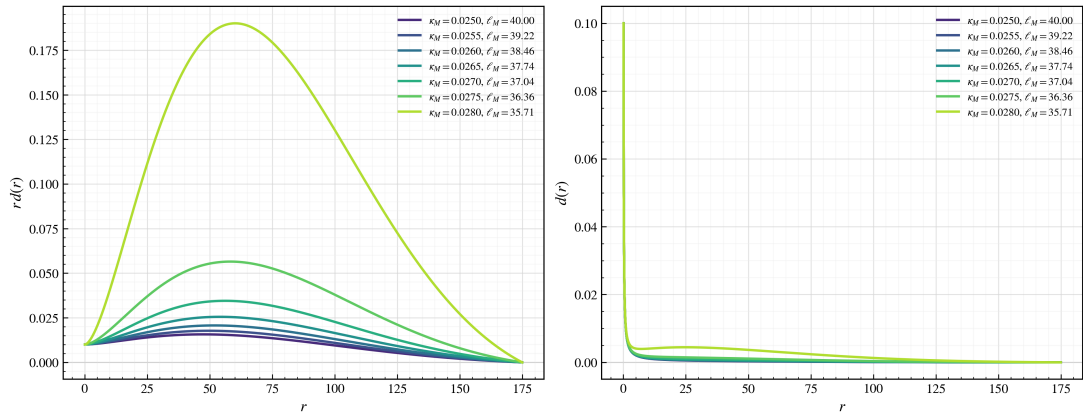
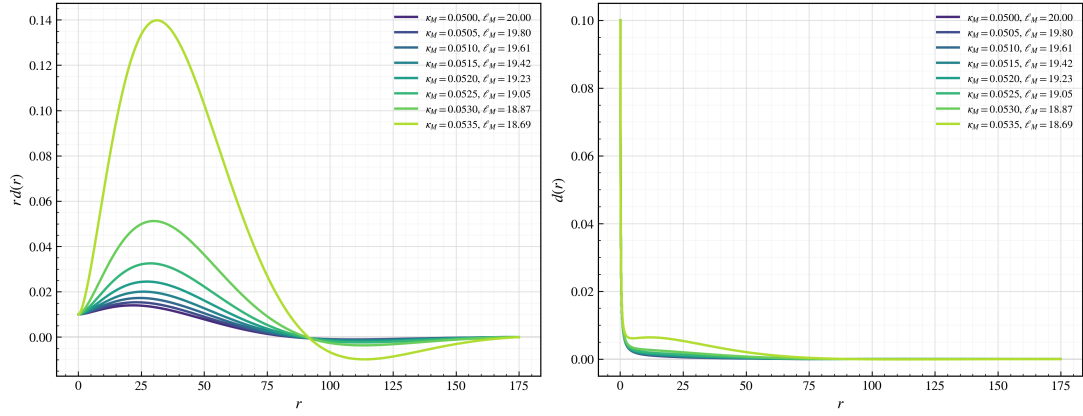

 (a) $\sin, L/\lambda \sim 1/2$

 (b) $\sin, L/\lambda \sim 1$

FIG. 3.1 Solution curves of Eq. (3.5) for different κ_M . Here L is $r_{\text{out}} - r_{\text{in}}$, and λ denotes the effective wavelength of the solution, whose profile resembles trigonometric functions. Note that the left panels display rd_r versus r , while the right panels display d_r versus r .

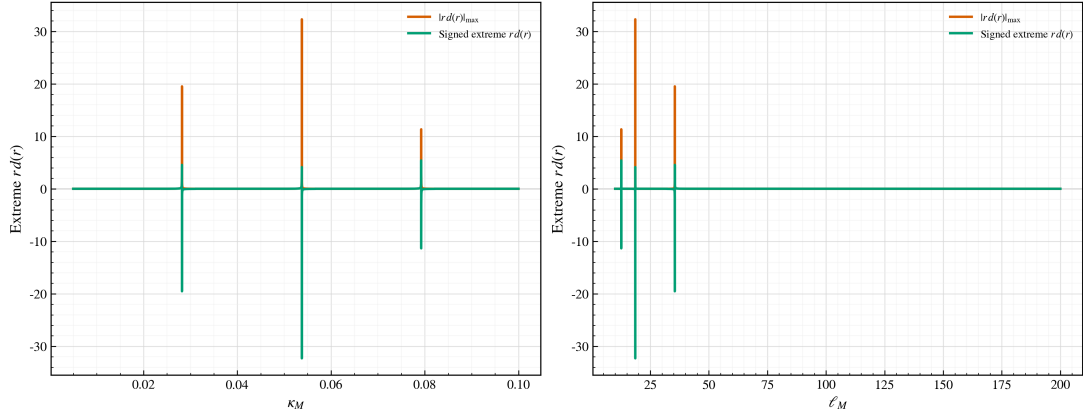
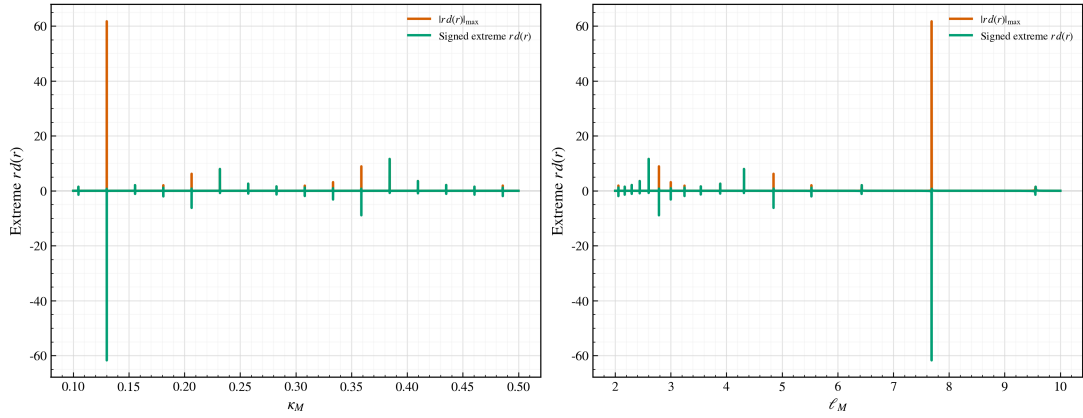
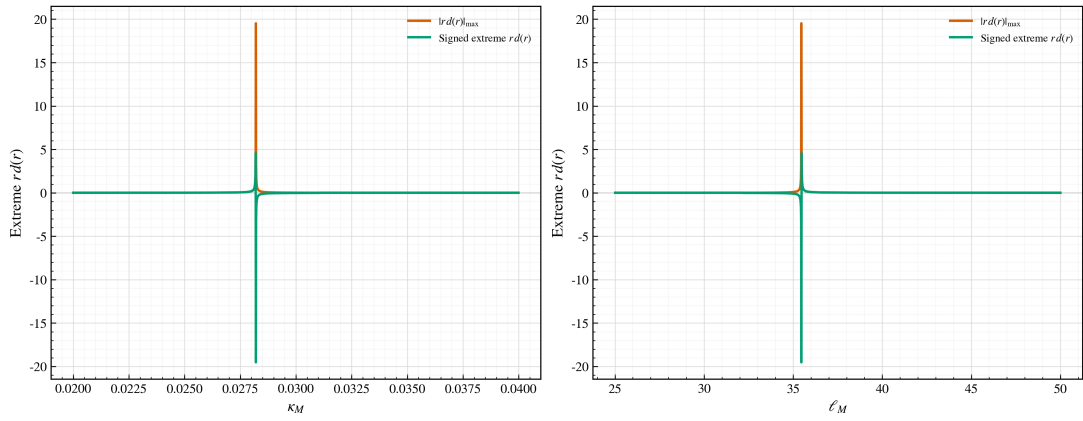
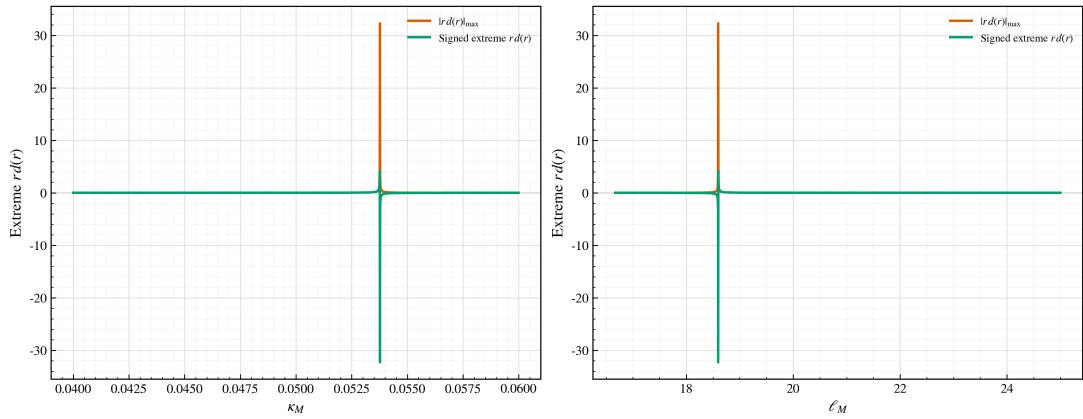

 (a) κ_M in (0.005, 0.1), ℓ_M in (10, 200)

 (b) κ_M in (0.1, 0.5), ℓ_M in (2, 10)

 FIG. 3.2 Maximum $|rd_r|$ and rd_r plotted against κ_M (left) and ℓ_M (right). A clear divergence of $|rd_r|$ and rd_r emerges at critical values of κ_M .

d_r itself emphasizes the decay of the displacement amplitude. Notably, the solution exhibits critical behavior at characteristic values of κ_M as L/λ varies, as shown in Fig. 3.2 (for a broad range of κ_M) and Fig. 3.3 (for a narrow range of κ_M). Additional detailed plots are provided in Appendix C. These critical values arise when the denominator in Eq. (3.5) becomes very small. They should therefore be understood as mathematical resonances of the boundary-value problem rather than as new material parameters. From the asymptotic behavior Eq. (3.6), we can expect the normalized monopole-screening displacement to satisfy $\sqrt{r}e^{\kappa_M r/\sqrt{2}}d_r(r) \sim O(1)\cos$, whereas the dipole-screening solution Eq. (1.13) follows $\sqrt{r}d_r(r) \sim O(1)\cos$. Profiles of the normalized quantity $\sqrt{r}e^{\kappa_M r/\sqrt{2}}d_r(r)$ for different κ_M are presented in Fig. 3.4. These theoretical features are observed as well in the fitting results shown in the next chapter. They provide a practical diagnostic: exponential decay after the $\sqrt{r}$ normalization points to monopole screen-



(a) Critical $\kappa_M = 0.02820$ and $\ell_M = 35.64$



(b) Critical $\kappa_M = 0.05377$ and $\ell_M = 18.60$

FIG. 3.3 Critical values of κ_M and ℓ_M for different wavenumber regimes.

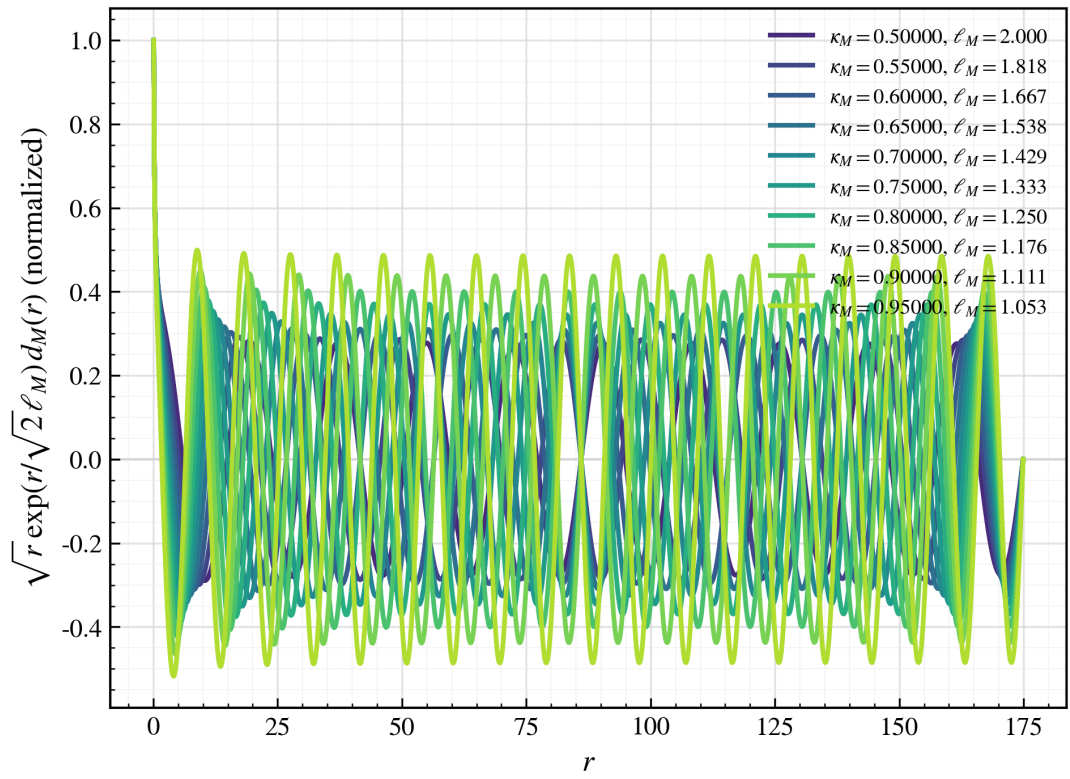


FIG. 3.4 Curves of $\sqrt{r} \exp(\kappa_M r / \sqrt{2}) d_r$. The quantity oscillates with nearly constant amplitude, consistent with theoretical expectations.



ing, while nearly constant-amplitude oscillation after the exponential compensation confirms the asymptotic structure of Eq. (3.6).



4 Numerical simulations and physical interpretation

To investigate effects of monopole screening within the pre-jamming phase, we study granular systems with the annular geometry illustrated in Fig. 1.3, a standard setup for elasticity studies of amorphous solids. In this chapter, we briefly introduce our numerical simulation setup, present key fitting results, and extract the physical features of monopole-screening behavior in the pre-jamming regime.

4.1 Data obtained from LAMMPS

Data for monopoles in a pre-jamming regime are generated via molecular-dynamics simulations performed using the LAMMPS package. Specifically, we employ two particle species of distinct sizes to suppress crystallization within a two-dimensional annular domain bounded by inner and outer radii $r_{\text{in}} \approx 1$ and $r_{\text{out}} \approx 172$. Interactions between the two particle types follow the Hertzian contact force, the standard contact-force model for granular systems, compared with other interaction potentials used for amorphous materials (glasses, colloids, etc.). This choice is suitable for the present problem because the particles interact only when they come into contact, as in dry granular packings. An inflation protocol is then applied to the inner boundary while the outer boundary remains fixed, consistent with the boundary conditions adopted for the analytical solutions Eq. (1.12) and Eq. (3.5). Here, the inner-boundary inflation is realized by expanding the central bigger particle inside the disk. The inflation process follows the athermal quasi-static (AQS) protocol, which proceeds sufficiently slowly to ensure validity of the equilibrium equations without irrelevant dynamic effects not considered in the thesis. The inner-boundary displacement d_0 appearing in analytical solutions for dipole screening and monopole screening is theoretically set as 10% of the large-particle radius $r = 1$, comparable to r_{in} , yielding $d_0 = 0.1$. In practice, however, inflation values of d_0 are around 0.1 yet frequently smaller in our numerical simulations, due to discrete-particle effects absent from the continuum-limit theory derived in Chapter 2. We therefore treat d_0 as a fitting parameter along with the typical length scales ℓ_P or ℓ_M .

We experimentally define a Scaling Factor (SF) linked to the particle packing fraction ϕ of



particles within the disk. At the jamming point, $SF=1.0$ corresponds to the jamming density $\phi_J = 0.842$. In the pre-jamming phase, we consider packing fractions ϕ as 0.7129, 0.7301, 0.7602, 0.7926, 0.8090, 0.8173, 0.8256, 0.8339, 0.8390, 0.8407, 0.8410, with corresponding SFs defined as 0.92, 0.931, 0.95, 0.97, 0.98, 0.985, 0.99, 0.995, 0.998, 0.999, 0.9992, respectively. These values allow us to track the approach to jamming from below over a broad range of unjammed states. We anticipate that dipole contributions grow while monopole contributions vanish as the system approaches the jamming point ϕ_J from lower ϕ .

In numerical simulations, we use 200 independent samples (labelled by running index, e.g. running 1, running 2, etc.) to obtain robust fitting outcomes and physical properties, identified by random numbers during the AQS inflation process. The use of many realizations is important because the geometric theory is most naturally interpreted at the level of individual samples, whereas averaged data are useful for extracting systematic trends.

4.2 Data Fitting

The domination of monopoles in the pre-jamming phase is demonstrated in this section. We retain 195 samples out of the initial 200 samples by discarding corrupted datasets. To simplify the data analysis, we use only the angular-averaged radial displacement field $\mathbf{d}(r, \theta) = \mathbf{d}(r)$, consistent with Eq. (3.5) and its assumptions. The averaged radial displacement field $d_r(r)$ is computed for concentric radial layers within the annulus, while the averaged angular displacement field $d_\theta(r)$ is approximately negligible. In all figures of this chapter, blue markers denote positive data points and pink markers denote negative ones.

We adopt the standard least-squares fitting approach equipped with grid-search, pre-fitting, and weighted-fitting routines for both dipole Eq. (1.12) and monopole Eq. (3.5), denoted as $d_{r,\text{dipole}}$ and $d_{r,\text{monopole}}$ respectively. Fitting is performed with respect to the scaled dependent variable rd_r rather than d_r , which improves the reliability and precision of the extracted fitting parameters. This scaling makes the tail behavior more visible and reduces the dominance of the largest displacement values near the inner boundary.

Additional supplementary figures are provided in Appendix C.

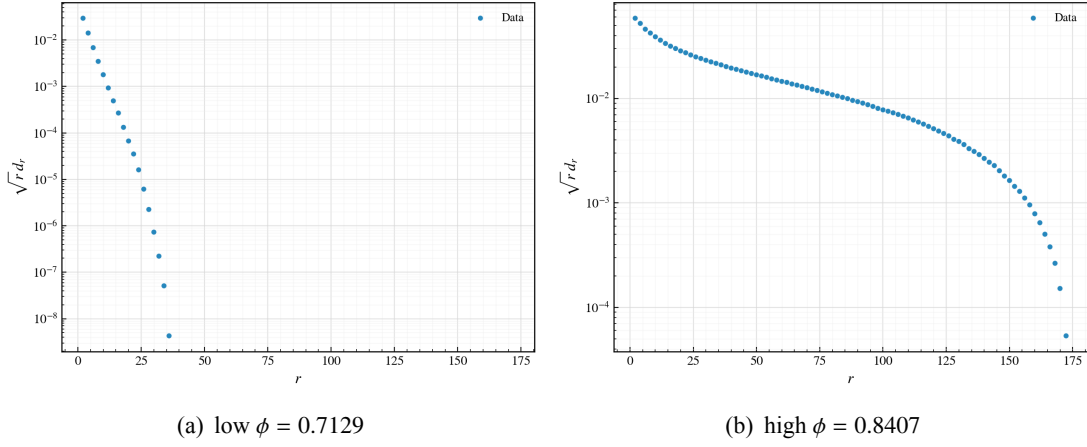


FIG. 4.1 Log-linear plots of the averaged scaled radial displacement field $\sqrt{r}d_r$. Based on the asymptotic forms Eq. (3.6) and Eq. (1.13), we predict a linear trend in the intermediate radial region r ($\ell_M \ll r \ll r_{out}$) for monopole screening, whereas no such linear signature appears for dipole screening. Note that $d_r = 0$ is numerically exact for $r \gtrsim 30$ of low ϕ (left panel), arising from the finite precision in simulations.

4.2.1 Averaged results over samples

We average the radial displacement field d_r across all 195 samples (running 1-200), with representative results displayed in Fig. 4.1 and Appendix C. Fitting outcomes of the radial displacement field d_r for both dipole and monopole screening are presented in Fig. 4.2 and Appendix C. Monopole-screening fits using Eq. (3.5) clearly outperform dipole-screening fits using Eq. (1.12). As the system approaches the jamming point, the fitting results of dipole screening gradually improve, consistent with our theoretical expectations. The superior agreement of monopole-screening fits originates from the exponential decay term in Eq. (3.6). Nevertheless, the fitted radial displacement field d_r exhibits an unusual negative region at small radii near r , while raw simulation data remain strictly positive. This negative feature arises from the oscillatory sine term in the asymptotic monopole solution Eq. (3.6), which predicts radial oscillations in theory. This mismatch is primarily from two effects: ensemble averaging and nonlinearity. First, averaging across multiple samples suppresses oscillations, recalling the application of the geometric theory discussed in Chapter 2. Second, nonlinear corrections become dominant far from jamming, which we address in the final section of this chapter.

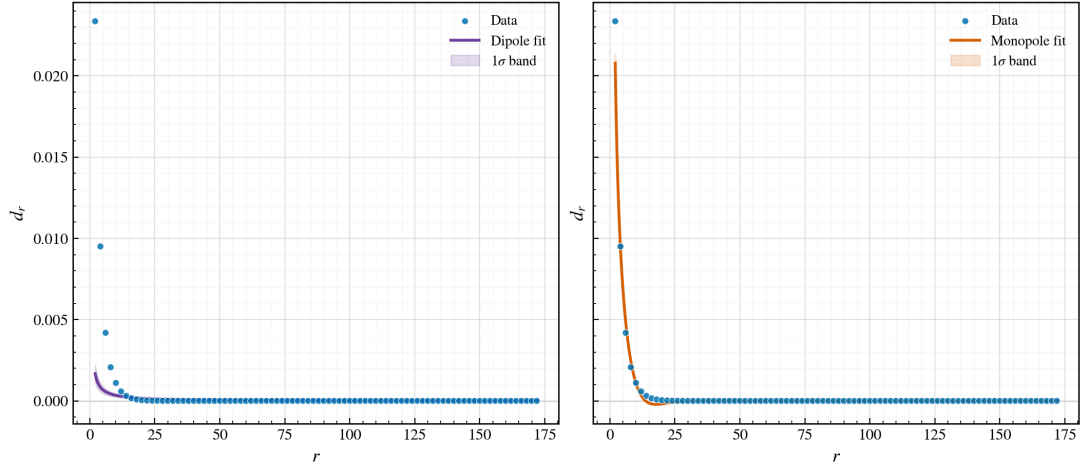
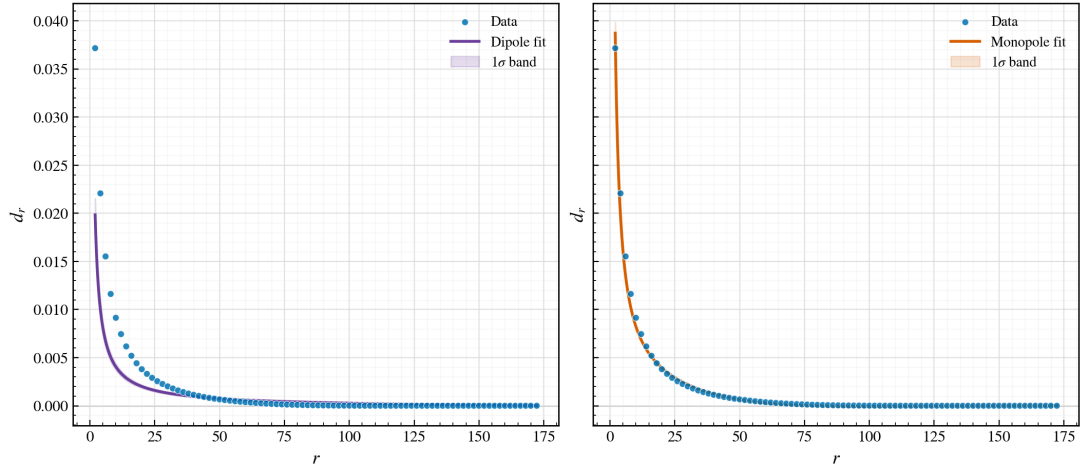
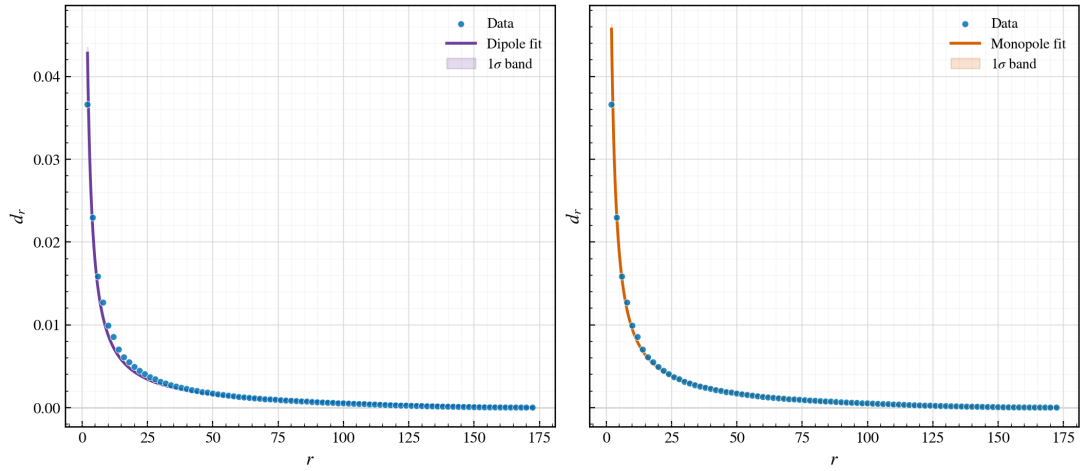

 (a) $\phi = 0.7301$: $\ell_M = 3.975$

 (b) $\phi = 0.8256$: $\ell_M = 21.230$

 (c) $\phi = 0.8410$: $\ell_M = 48.530$

FIG. 4.2 Left panels: fitting results of dipole screening. Right panels: fitting results of monopole screening. Both are obtained via identical fitting protocols. Note that the unmentioned typical length scale for dipoles ℓ_P is excessively large ($> 10^9$), making it physically irrelevant within the unified screening equation Eq. (2.24).

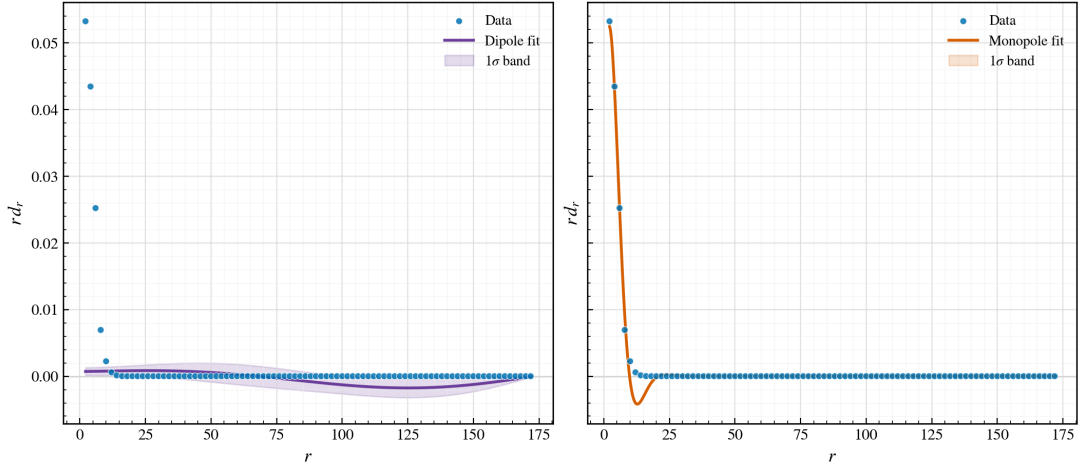
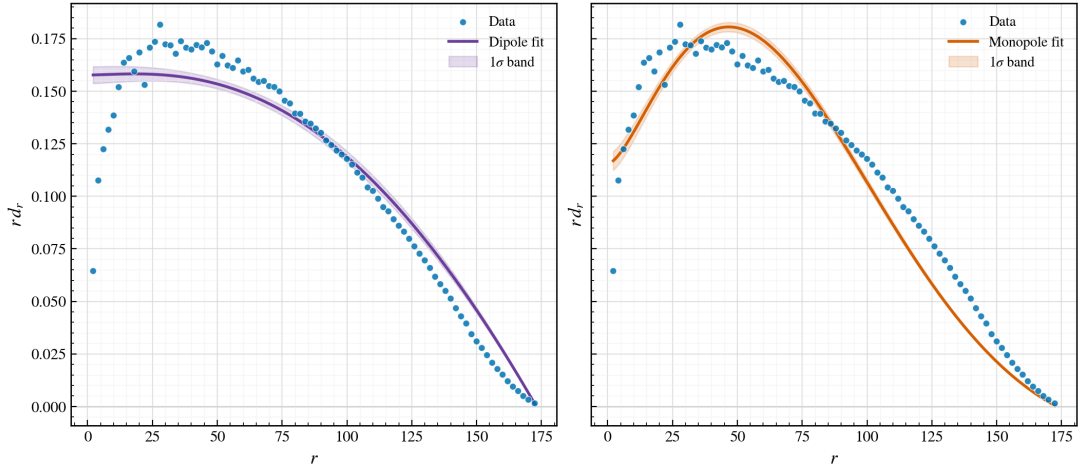

 (a) low $\phi = 0.7129$ (running 81): $\ell_M = 2.446$

 (b) high $\phi = 0.8407$ (running 23): $\ell_M = 39.600$

FIG. 4.3 Two single-sample fits at low and high packing fractions. Raw data remain strictly positive. Note that the unmentioned typical length scale for dipoles ℓ_P is excessively large ($> 10^9$) and physically negligible.

4.2.2 Single samples for each running

Since the geometric theory and derived solution Eq. (3.5) apply most directly to individual physical realizations, we perform separate fits for each single sample. Two representative single-sample results are shown in Fig. 4.3. Both favor effects of monopole screening: the low- ϕ case displays a negative fitted region at small r as well, while the high- ϕ case features an upward tail in rd_r at small r , which outperforms the dipole one. These single-sample signatures are consistent with the averaged ones in Fig. 4.2. Three representative oscillatory single-sample profiles are shown in Fig. 4.4. Interestingly, the fitting curves rd_r of monopole screening exhibit

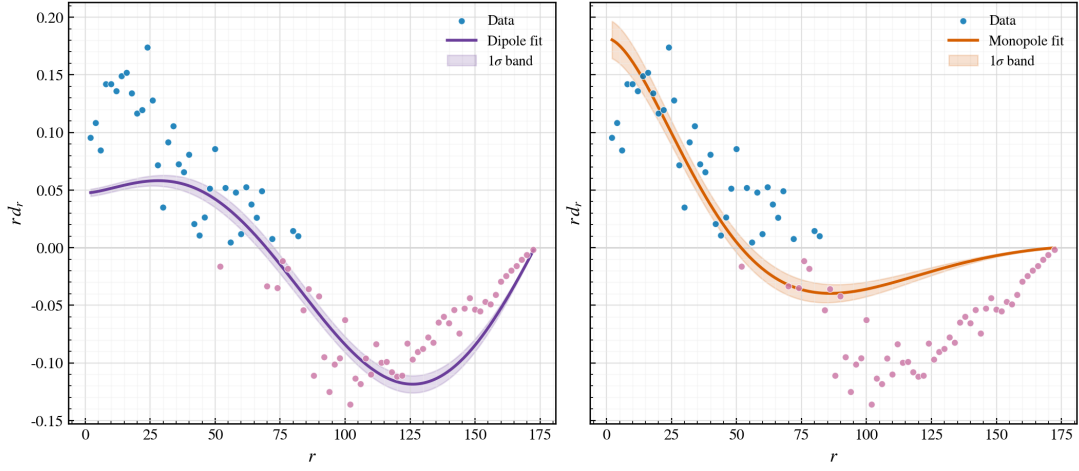
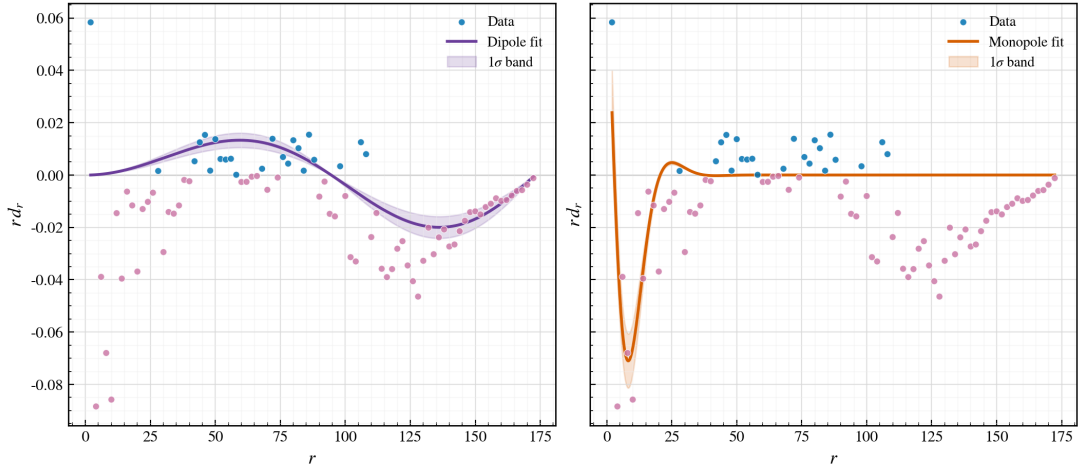
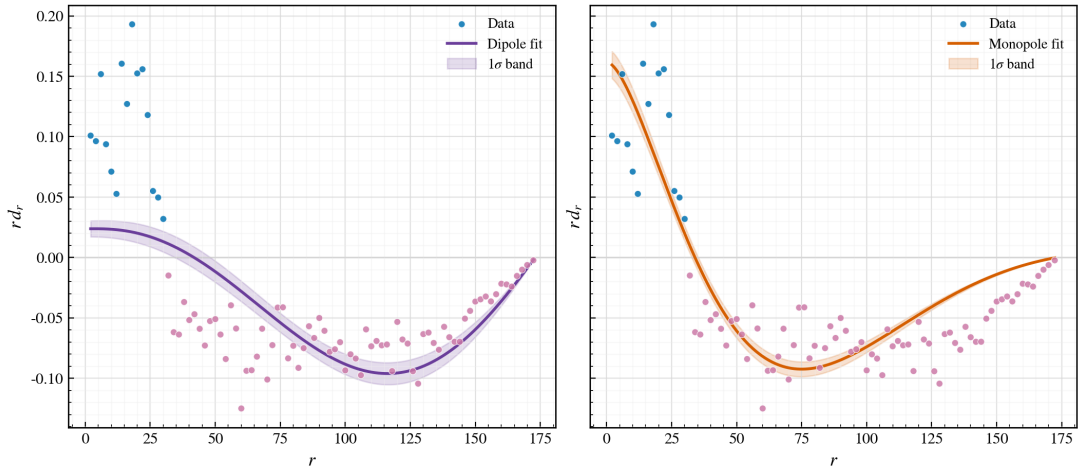

 (a) oscillation 1 (running 165): $\ell_P = 31.956, \ell_M = 26.752$

 (b) oscillation 2 (running 157): $\ell_P = 24.641, \ell_M = 3.463$

 (c) oscillation 3 (running 196): $\ell_P = 39.239, \ell_M = 30.295$

FIG. 4.4 Oscillatory displacement fields with $\phi = 0.8410$, the packing fraction closest to the jamming point $\phi_J = 0.8420$ in our simulations. For numerical simulations, oscillations commonly emerge near jamming due to particle rebounds upon collision with the outer boundary, and become more pronounced at higher packing fractions within the pre-jamming regime. Characteristic length scales associated with such boundary-induced rebounds are discussed later.

pronounced oscillations near the jamming point, a feature poorly captured by dipole-screening or purely elastic solutions, especially at small radii. It is natural to expect the contribution of dipole screening as the complement of the monopole one. We therefore adopt a linear superposition of dipole- and monopole-screening contributions, instead of solving Eq. (2.24) directly, to describe such oscillatory behavior, with the phenomenological combined fitting function

$$d_r(r) = B d_{r,\text{dipole}}(r) + (1 - B) d_{r,\text{monopole}}(r), \quad (4.1)$$

where the mixing weight B satisfies $B \in (0, 1)$ and the inner-boundary displacement d_0 is identical for both components. Fits using this combined ansatz Eq. (4.1) for the three oscillatory samples are shown in Fig. 4.5, strongly suggesting that displacement fields near the jamming point arise from direct superposition of dipole- and monopole-screening effects. The combined dipole-monopole ansatz Eq. (4.1) yields nearly perfect fits compared with individual models of dipole and monopole screening (Fig. 4.4), which confirms our proposed superposition mechanism near the jamming transition.

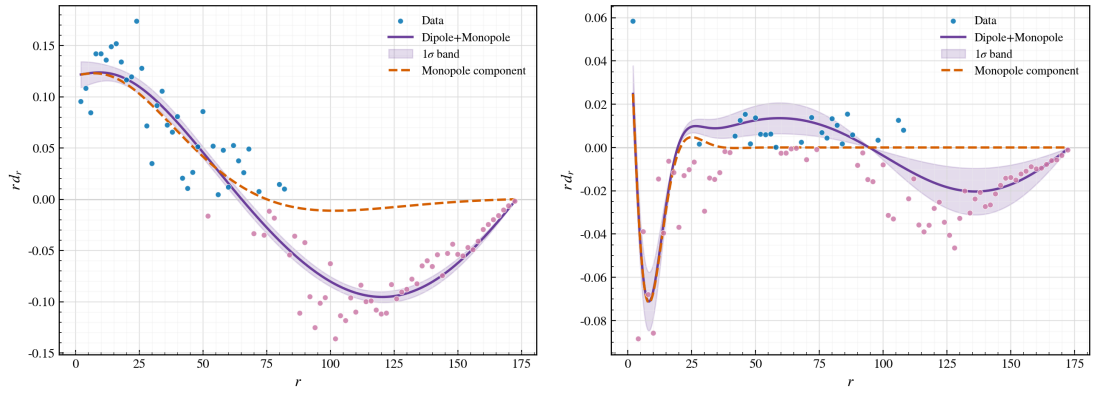
4.3 Length scales

Physical behaviors and characteristics in the pre-jamming phase are dominated primarily by effects of monopole screening. Length scales are critical to understanding the underlying physics. In this section, we define three distinct length scales and demonstrate their consistency based on our simulation data. Although these length scales are obtained in different ways, they describe related aspects of the same screened displacement response.

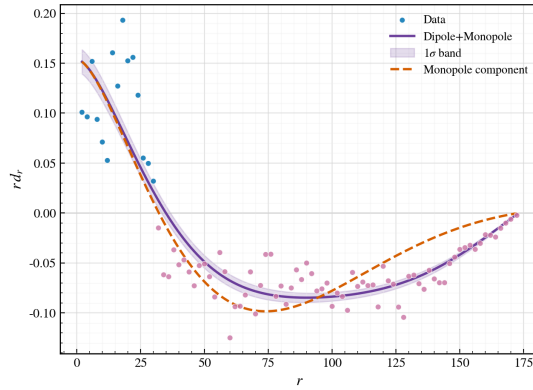
4.3.1 Definitions

Typical/Interaction Length Scale ℓ_M : This type of length scales is defined in the particular solutions Eq. (3.5), which is first derived in the screening equation Eq. (2.23) and defined in the coupling constant Eq. (2.9) of monopoles. It quantifies the strength of monopole-monopole interactions. Another related length scale for monopole screening is the length scale $L_M = \sqrt{2}\ell_M$ as in Eq. (3.6).

Correlation Length Scale ξ : ξ is defined as $r - r_{\text{in}}$ where $r \in (r_{\text{in}}, r_{\text{out}})$ is the smallest radial coordinate (first point on the horizontal axis) satisfying $d_r(r = \xi) = d_{\text{th}} = 10^{-3}$. Different



(a) oscillation 1 (running 165): $\ell_P = 34.551, \ell_M = 21.827, B = 0.284529$ (b) oscillation 2 (running 157): $\ell_P = 24.641, \ell_M = 3.807, B = 0.000012$



(c) oscillation 3 (running 196): $\ell_P = 24.640, \ell_M = 30.635, B = 0.000002$

FIG. 4.5 Combined dipole-monopole fits (Eq. (4.1)) for oscillatory profiles at $\phi = 0.8410$ with free parameters d_0, ℓ_P, ℓ_M and B . Panel (b) and panel (c) are almost dominated by monopole screening alone with large errors of B .

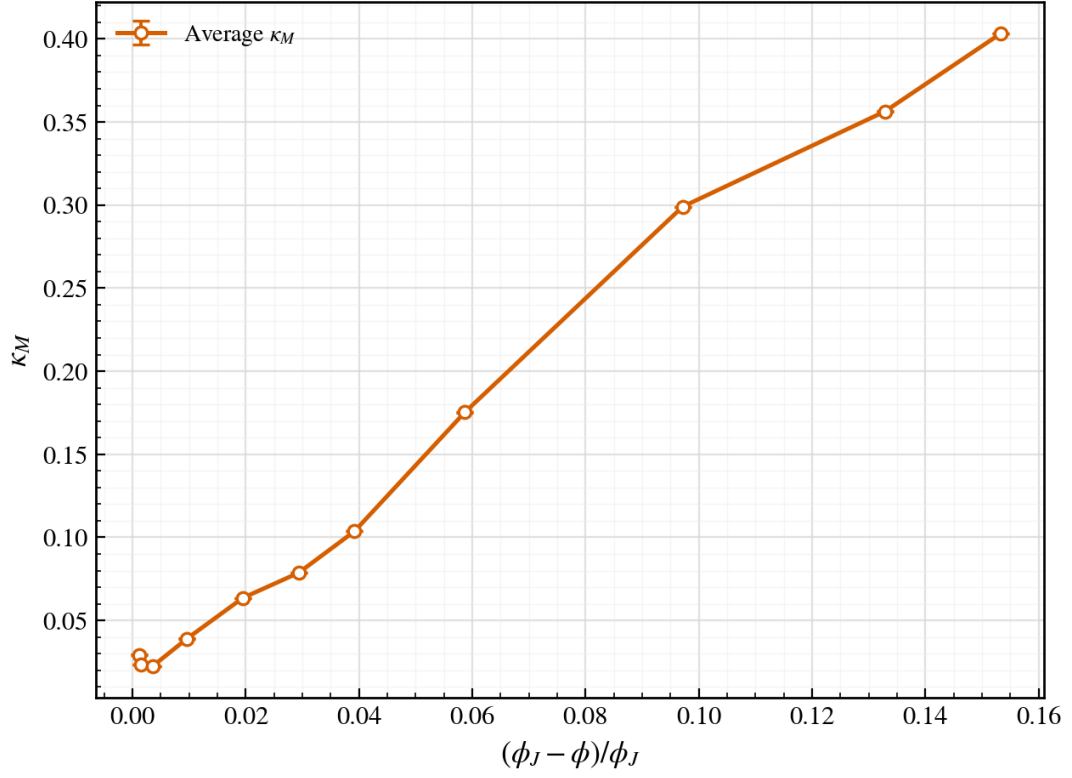
threshold values d_{th} produce distinct curves, while their slopes are approximately identical. This length scale describes the spatial correlation length of radial displacement fields and the rate of their decay, and it is confirmed to diverge as given by Eq. (1.4) in the thermodynamic limit.

Decay Length Scale R_0 : Similar to the correlation length scale ξ , R_0 is defined as $r - r_{\text{in}}$. Here, $r \in (r_{\text{in}}, r_{\text{out}})$ is the smallest radial coordinate (first point on the horizontal axis) satisfying $d_r(r = R_0) = 0$ in the absence of pronounced oscillations, otherwise r is set to r_{out} . It quantifies the spatial range of monopole-screening influence.

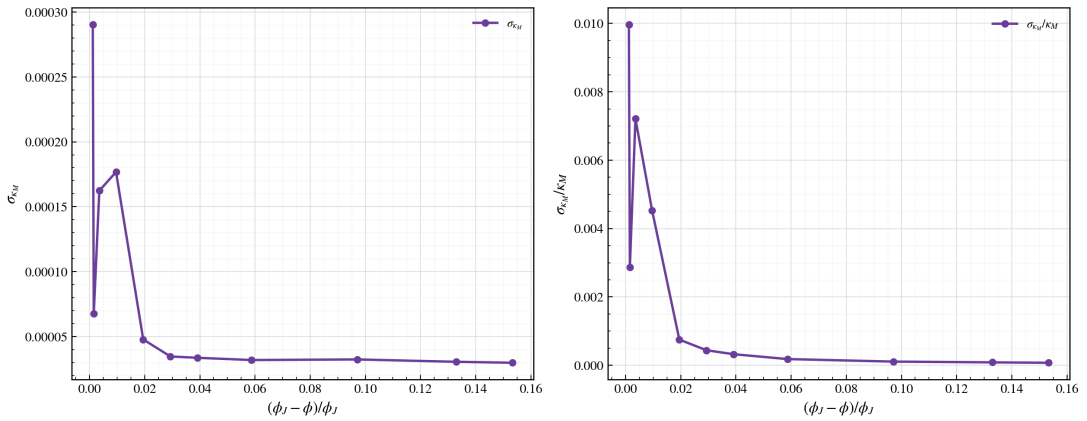
4.3.2 Wave number and different length scales

In this subsection, we first fit individual single samples and then average the results to obtain precise length scales and stable slopes, as we discussed before. To illustrate the tendency of the wave number κ_M and its corresponding typical length scale with respect to the packing fraction ϕ , we plot κ_M against the renormalized packing fraction $(\phi_J - \phi)/\phi_J$ in Fig. 4.6. We find that tiny fluctuations of κ_M or ℓ_M grow as the system approaches the jamming point.

The length scales ℓ_M (corresponding to κ_M), ξ and R_0 are plotted together against $(\phi_J - \phi)/\phi_J$ in Fig. 4.7. Note that we plot $L_M = \sqrt{2}\ell_M$ instead of ℓ_M itself, consistent with the asymptotic expansion Eq. (3.6). The key reason is that L_M governs the exponential decay behavior of the displacement field. Here, the difference of slopes between three length scales as $\phi_J - \phi \approx 0$ is because of the limited system size, namely limited particle number N [17]. While R_0 maintains the trend, finite size effects suppress the divergence of ξ and L_M as the system approaches the jamming point. The collected graphs Fig. 4.7 of length scales shows: First, effects of monopole screening dominates due to $L_M \ll \ell_P$, see Fig. 4.2 and Fig. 4.3; Second, the density of monopoles in numerical samples is inferred to be almost uniform because $R_0 \propto \xi \propto L_M$ roughly. Given the divergent scaling law of ξ in Eq. (1.4), a log-log plot of ξ versus $(\phi_J - \phi)$ should yield a linear relationship with slope $-\nu$, consistent with the right panels of Fig. 4.7. To calculate the scaling exponent ν , we perform a linear fit of ξ in Fig. 4.8, excluding the first 5 data points (those closest to ϕ_J) to ensure stability. Note that the fitting choice of different data points and methods only slightly modifies the exact value ν and does not affect the overall conclusion. Fig. 4.8 verifies the scaling law $\xi \sim (\phi - \phi_J)^{-\nu}$ with the scaling exponent $\nu \approx 0.8$ (ranging from 0.7 to 0.9 according to fitting choices).



(a) κ_M vs $(\phi_J - \phi)/\phi_J$



(b) σ_{κ_M} vs $(\phi_J - \phi)/\phi_J$

FIG. 4.6 Panel (a) shows the average value of κ_M as a function of $(\phi_J - \phi)/\phi_J$. Panel (b) shows the standard deviation and relative error of κ_M as a function of $(\phi_J - \phi)/\phi_J$. Note that fluctuation of κ_M or ℓ_M grows as the system approaches the jamming point.

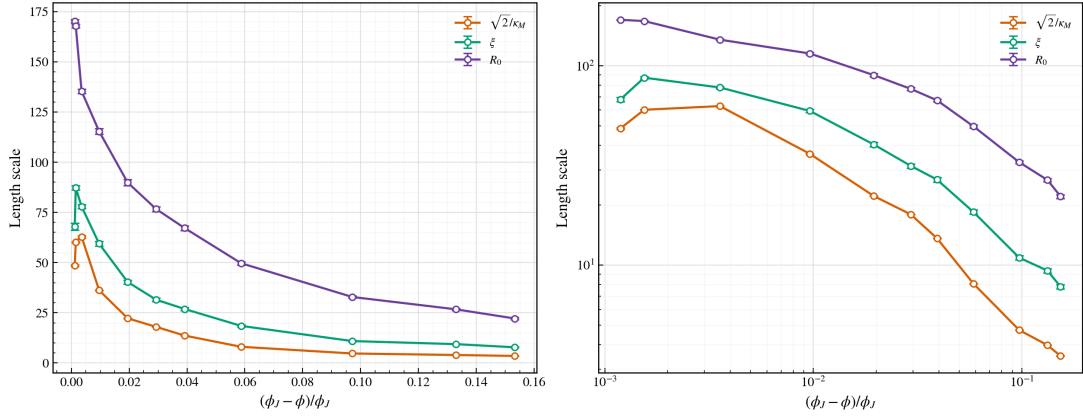
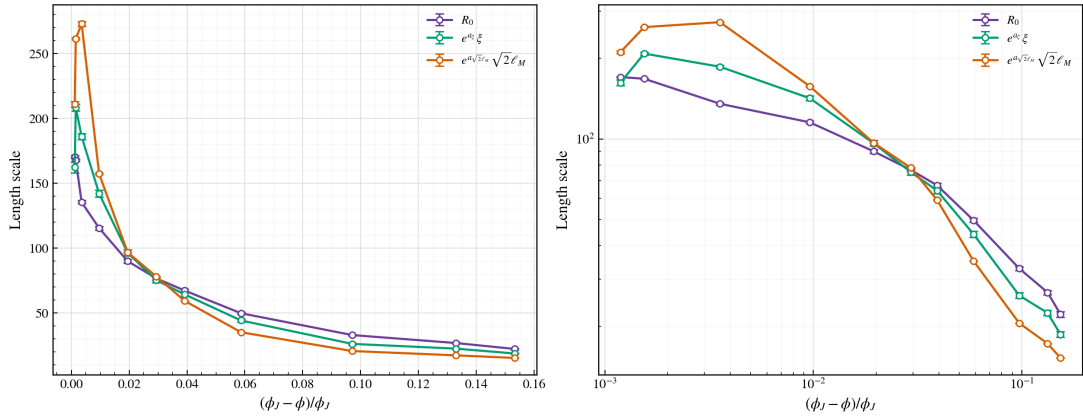

 (a) Length scales vs $(\phi_J - \phi)/\phi_J$

 (b) Collapsed Length scales vs $(\phi_J - \phi)/\phi_J$

FIG. 4.7 Panel (a) shows the length scales ℓ_M , ξ and R_0 as function of $(\phi_J - \phi)/\phi_J$ in linear and log-log plot. Panel (b) shows the length scales after shifting (collapsing) as function of $(\phi_J - \phi)/\phi_J$ to clarify their trends and their properties in linear and log-log plot. All three length scales, interaction length scale ℓ_M , correlation length scale ξ and decay length scale R_0 , exhibit the same qualitative trend. Differences in their slopes near the jamming point scales arise from finite size effects, i.e., a limited number of particles N .

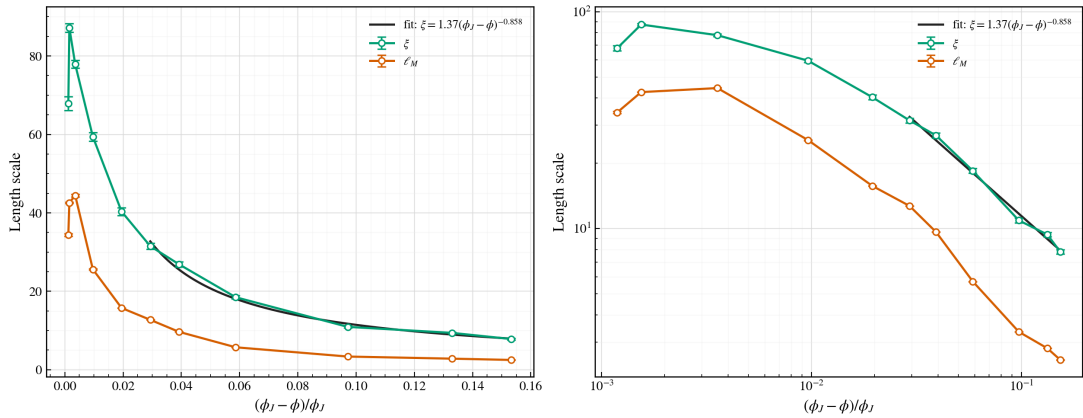


FIG. 4.8 Length scales ξ and ℓ_M as functions of $(\phi_J - \phi)/\phi_J$ in linear (left panel) and log-log (right panel) plots. Here, we fit the scaling law with excluding the first 5 data points (closest to ϕ_J) to ensure stability.



4.4 Discussion

In this chapter, we have illustrated that effects of monopole screening dominate in the pre-jamming phase (Fig. 4.1, Fig. 4.2 and Fig. 4.3). As the system approaches the jamming point, effects of dipole and monopole screening combine cooperatively (Fig. 4.4 and Fig. 4.5). We further define three length scales (interaction length scale ℓ_M , correlation length scale ξ and decay length scale R_0) to quantify key physical features of amorphous solids observed in our numerical simulations. Fitting results for these length scales indicate an approximately uniform distribution of topological charges (monopoles) and verify the divergent scaling law of ξ in Eq. (1.4).

However, as noted in Fig. 4.2 and Fig. 4.3, the fitted radial displacement field d_r exhibits an unusual negative region at small radii, especially for low packing fractions ϕ far from the jamming point within the unjammed phase. This discrepancy originates primarily from nonlinear corrections neglected in the analytical solution Eq. (3.5). Within the geometric framework, the Gaussian-curvature equation Eq. (2.19) is derived from the exact geometric (metric) equation Eq. (2.17) under linearized approximations, both assuming sufficiently low quadrupole density Q . This assumption fails for more fluid-like configurations corresponding to low packing fractions ϕ below ϕ_J . Moreover, finite size effects become significant near the jamming point [17], as shown in Fig. 4.7.

To address the frequent oscillatory features observed in high ϕ samples, we should revisit the boundary conditions imposed for the particular solution Eq. (3.5). As ϕ approaches ϕ_J and L_M becomes comparable to the system size at the same time, the divergence-free boundary condition $d_r(r) \rightarrow 0 (\ell_M \ll r \ll r_{\text{out}})$ probably breaks down in a few samples, permitting non-vanishing contributions from the divergent terms ber_1 and bei_1 . These first-order Kelvin functions ber_1 and bei_1 produce obvious oscillations at large radial distances. Even with this violation of boundary conditions, the solution (3.5) remains the optimal fitting function from a numerical perspective.

An important feature emerging from our numerical simulations is the coexistence of dipole-screening and monopole-screening behaviors near the jamming point. While monopole screening dominates clearly in low- ϕ unjammed systems, dipole contributions become increasingly relevant as ϕ approaches ϕ_J . Physically, this coexistence reflects the crossover nature of the jam-



ming transition itself. Far below the jamming point, the system behaves similarly to a highly disordered fluid-like state. Mechanical perturbations are rapidly screened, resulting in strongly damped displacement fields governed primarily by monopole screening. In contrast, near the jamming point, the system gradually develops collective rigidity and enhanced spatial correlations. The displacement field can then partially recover elastic-like propagation, leading to the reappearance of oscillatory dipole-screening contributions. From this perspective, the jamming transition should not be interpreted as an abrupt replacement of monopole screening by dipole screening. Instead, the two mechanisms coexist and compete within a crossover regime close to ϕ_J . The combined fitting ansatz Eq. (4.1) therefore provides a natural physical interpretation of the displacement fields observed numerically. The fitting parameter B may be interpreted phenomenologically as a measure of the relative importance of elastic-like dipole propagation compared with strongly screened monopole responses. Moreover, the oscillatory profiles observed near jamming are likely enhanced further by finite-size effects and boundary reflections.



Conclusion

In this thesis, we investigate the effects of monopole screening in disordered solids, focusing on a two-dimensional disk of granular systems illustrated in Fig. 1.3. We derive the monopole screening equation Eq. (3.1) from the geometric theory of anomalous elasticity and analyze its mathematical structure and key features. The theoretical construction starts from the metric description of amorphous solids and connects plastic events with effective geometric charges. Within this framework, quadrupoles, dipoles and monopoles appear as different levels of the same multipole hierarchy, corresponding respectively to quasielastic behavior, plastic-solid behavior and the pre-jamming regime.

The analytical radial displacement field for monopoles solved from the corresponding screening equation reads Eq. (3.5) effectively, whose asymptotic behavior Eq. (3.6) features an exponential-decay term that distinguishes monopole behavior fundamentally from effects of quadrupole renormalization and dipole screening. This exponential factor introduces a finite screening length and shows that mechanical perturbations are strongly suppressed in the monopole-dominated regime. The solution also contains an oscillatory factor inherited from Kelvin functions and the finite annular boundary conditions. These mathematical features provide the basic criteria used later to interpret numerical displacement fields.

Interestingly, we demonstrate that effects of monopole screening dominate the mechanical behavior in the pre-jamming phase. This complements existing results, where the quasielastic phase is governed by quadrupole renormalization and the plastic solid phase by dipole screening. Near the jamming transition, numerical displacement fields further suggest coexistence and competition between dipole propagation and monopole screening, reflecting the crossover nature of mechanical dipole- and monopole-screening effects. We further define three distinct length scales and verify their consistency via numerical simulations. These length scales grow and ultimately diverge as the system approaches the jamming point, accompanied by a reduction in quadrupole density. The agreement between the interaction length scale, the correlation length scale and the decay length scale suggests that monopoles are approximately uniformly distributed in the simulated samples.



Several aspects remain open for future improvement. First, nonlinearity and interactions between different types of topological charges are neglected in the present geometric framework, although they become important for the regimes studied in this work. We hypothesize that these contributions govern the physics of the unjammed regime far from jamming. Second, the boundary conditions for monopole screening may require improvements. They can vary with numerical schemes and sample-preparation protocols for higher precision, with far more complexity. In particular, the oscillatory samples close to jamming indicate that the finite outer boundary and possible divergent Kelvin-function components may need to be treated more carefully. Finally, the geometric theory used here describes individual realizations more naturally than ensemble-averaged data. Future work should therefore combine nonlinear geometric corrections, inter-type charge interactions and more detailed sample-level statistics. We anticipate that these advanced corrections can be incorporated into future studies.



Acknowledgement

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回首在北航四年的本科岁月，如今我已是二十二岁，刚入学时的场景依旧清晰难忘——年纪增长，反倒越发觉得时间过得悄然无声。我常常思索，自己为何坚持学习物理，又为何选择凝聚态理论方向。此前蔡子老师也曾问过我这个问题，当时我没能给出明确的答案。如今再来回答，想法和从前相差不大，我始终觉得不必刻意深究缘由：“纠结这个问题纯度太低了！我从来没想过这个问题。”既然已经走上这条路，我更愿意安心享受钻研学习的过程，一味纠结初衷，反而没有太大意义。经过四年的学习，我慢慢确定了自己的研究方向。首先十分感谢智朝晖学长、崔健老师，还有刘雷轶男师兄以及北航量子信息学习小组的各位同学，在一次次交流探讨中，我找准了自己的 taste，最终确定投身凝聚态理论研究。同时感谢物理学院张其安老师、空间学院刘焱老师，两位老师分别带我了解了格点 QCD、黑洞热力学相关知识，让我积累了不少有趣的科研实践经验。感谢本科大创指导老师魏光美老师，还有苗澍、印子一、潘晨阳几位一同合作的同学。也感谢李嘉航、刘传、杨京华、詹骋昊等同学，与我一起探讨物理知识，互相学习进步。在准备保研的过程中，我有幸和上海交大蔡子老师、清华杨硕老师、复旦张鹏飞老师、浙大刘钊老师线上当面交流，也通过邮件向中科院卡弗里所朱征老师、清华汪忠老师请教问题，感谢各位老师都给予我的建议。学习之余，我还要特别感谢北航飞梦 ACG 联萌，尤其是飞梦动画同好会的朋友们，陪我度过轻松愉快的四年大学时光。闲暇时原神、崩坏：星穹铁道、minecraft 等各类游戏、动漫以及喜欢的虚拟主播，丰富了我的课余生活；身边一众好友的陪伴，也让我的大学生活格外充实。最后，由衷感谢我的家人，一直以来默默支持我、信任我，做我最坚实的依靠，让我能够安心完成学业，追逐自己的目标。

“做不到想做的事也许才是正解，说不出想说的话也不是错误。”



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Appendix A Screening equations for displacement fields in planar geometry

It is natural to set $\bar{K}^0 = 0$ in a two-dimensional disk shown in Fig. 1.3. Under this planar approximation, the reference geometry is flat and the screening equations for the Airy stress function reduce to

$$\frac{1}{\bar{Y}} \Delta \Delta \chi = 0, \quad \text{Quadrupole;} \quad (\text{A.1})$$

$$\frac{1}{\bar{Y}} \Delta \Delta \chi + \frac{1}{\bar{Y}} \ell_P^{-2} \Delta \chi = 0, \quad \text{Dipole;} \quad (\text{A.2})$$

$$\frac{1}{\bar{Y}} \Delta \Delta \chi + \frac{1}{\bar{Y}} \ell_M^{-4} \chi = 0, \quad \text{Monopole.} \quad (\text{A.3})$$

To get the relation between the Airy stress function χ and the displacement field $\mathbf{d}$, we substitute Eq. (1.1) into Eq. (2.18) and rewrite Eq. (1.2) with $g = \delta$ and $\nabla = \partial$ in planar geometry:

$$\varepsilon_{\mu\gamma} \varepsilon_{\nu\delta} \sigma^{\mu\nu} = \varepsilon_{\mu\gamma} \varepsilon_{\nu\delta} \varepsilon^{\mu\alpha} \varepsilon^{\nu\beta} \partial_{\alpha\beta} \chi = \delta_\gamma^\alpha \delta_\delta^\beta \partial_{\alpha\beta} \chi = \partial_{\gamma\delta} \chi \Rightarrow \partial_{\alpha\beta} \chi = \varepsilon_{\mu\alpha} \varepsilon_{\nu\beta} \sigma^{\mu\nu}, \quad (\text{A.4})$$

$$g_{\alpha\beta} A^{\alpha\beta\gamma\delta} = 2\lambda g^{\gamma\delta} + 2\mu g^{\gamma\delta} = \lambda_1 g^{\gamma\delta}, \quad (\text{A.5})$$

where $\lambda_1 = 2\lambda + 2\mu$. Eq. (A.4) expresses the Airy stress representation in a flat metric, while Eq. (A.5) is the trace of the isotropic elastic tensor. So we can write further with Eq. (A.4) and Eq. (A.5):

$$\begin{aligned} \Delta \chi &= g^{\alpha\beta} \partial_{\alpha\beta} \chi = g^{\alpha\beta} \varepsilon_{\mu\alpha} \varepsilon_{\nu\beta} \sigma^{\mu\nu} \\ &= g^{\alpha\beta} \varepsilon_{\mu\alpha} \varepsilon_{\nu\beta} \sigma^{\mu\nu} = \varepsilon_\mu^\beta \varepsilon_{\nu\beta} A^{\mu\nu\gamma\delta} u_{\gamma\delta} \\ &= (g_{\mu\nu} g^\beta_\beta - g_{\mu\beta} g^\beta_\nu) A^{\mu\nu\gamma\delta} u_{\gamma\delta} \\ &= g_{\mu\nu} A^{\mu\nu\gamma\delta} \partial_{(\gamma} d_{\delta)} = \lambda_1 g^{\gamma\delta} \partial_\gamma d_\delta \\ &= \lambda_1 \nabla \cdot \mathbf{d}. \end{aligned}$$



Finally we obtain the relation equation

$$\Delta\chi = \tilde{Y}'\nabla \cdot \mathbf{d} \quad (\text{A.6})$$

where the effective Young's modulus is defined as $\tilde{Y}' = 2\lambda + 2\mu$. This identity is the bridge between the potential formulation and the displacement-field formulation used in the simulations.

A.1 Quadrupole

In the quadrupole regime, the screening equation Eq. (A.1) substituted Eq. (A.6) into becomes

$$\frac{\tilde{Y}'}{\tilde{Y}}\Delta(\nabla \cdot \mathbf{d}) = 0. \quad (\text{A.7})$$

After cancelling the factor and applying a ∇ , we obtain a bi-harmonic equation

$$\Delta\Delta\mathbf{d} = 0. \quad (\text{A.8})$$

This is the ordinary long-range elastic equation in the present formulation. The quadrupole contribution only renormalizes elastic constants, so no finite screening length appears at this level.

A.2 Dipole

In the dipole regime, the screening equation Eq. (A.2) substituted Eq. (A.6) into becomes

$$\frac{\tilde{Y}'}{\tilde{Y}}\Delta(\nabla \cdot \mathbf{d}) + \frac{\tilde{Y}'}{\tilde{Y}}\ell_p^{-2}\nabla \cdot \mathbf{d} = 0, \quad (\text{A.9})$$

$$\frac{\tilde{Y}'}{\tilde{Y}}\nabla \cdot (\Delta\mathbf{d} + \ell_p^{-2}\mathbf{d}) = 0. \quad (\text{A.10})$$

After cancelling the factor and applying a ∇ , we obtain a Helmholtz equation with gauges

$$\Delta\Delta\mathbf{d} + \ell_p^{-2}\Delta\mathbf{d} = 0. \quad (\text{A.11})$$



The additional $\ell_p^{-2}\Delta\mathbf{d}$ term introduces the characteristic dipole length scale. It changes the spatial structure of the displacement field, but still acts through derivatives of $\mathbf{d}$.

A.3 Monopole

In the monopole regime, the screening equation Eq. (A.3) substituted Eq. (A.6) into after applying a Δ becomes

$$\frac{\tilde{Y}'}{\tilde{Y}}\Delta\Delta(\nabla\cdot\mathbf{d}) + \frac{\tilde{Y}'}{\tilde{Y}}\ell_M^{-4}\nabla\cdot\mathbf{d} = 0, \quad (\text{A.12})$$

$$\frac{\tilde{Y}'}{\tilde{Y}}\nabla\cdot(\Delta\Delta\mathbf{d} + \ell_M^{-4}\mathbf{d}) = 0. \quad (\text{A.13})$$

After cancelling the factor, we obtain a bi-Helmholtz equation with a source term (the integration constant as a gauge-invariant quantity)

$$\Delta\Delta\mathbf{d} + \ell_M^{-4}\mathbf{d} = \mathbf{C}(\mathbf{r}), \quad (\text{A.14})$$

with the divergence-free condition $\nabla\cdot\mathbf{C} = 0$. The appearance of $\mathbf{C}(\mathbf{r})$ reflects the fact that the divergence equation determines the vector equation only up to a divergence-free field.

To analyze $\mathbf{C}(\mathbf{r})$, we consider a radially symmetric system where the displacement field is defined purely in the radial direction: $\mathbf{d}(r, \theta) = d_r(r)\hat{\mathbf{r}}$, and $\mathbf{C}(\mathbf{r}) = C(r)\hat{\mathbf{r}}$ with polar symmetry as well. From the divergence-free condition in polar coordinates, we can obtain the basic form

$$\frac{1}{r}\frac{\partial}{\partial r}(rC(r)) = 0 \Rightarrow C(r) = \frac{C_1}{r}, \quad (\text{A.15})$$

where the only permissible source term is the inverse radial term, representing a long-range source. The corresponding particular solution driven by this source is $d_r \sim \ell_M^4 C_1/r$. Since the decay $d_r \sim r^{-1}$ of the particular solution represents unscreened long-range behavior, which is just the behavior of the classical elasticity, it does not affect the monopole-screening physics to set $C_1 = 0$, namely $\mathbf{C}(\mathbf{r}) = 0$. With this gauge choice, the displacement-field equation used in the main text is obtained.



Appendix B Solutions of monopole screening equation

B.1 Three representations of general solutions

When the monopole screening dominates in the pre-jamming regime, the equation is Eq. (3.1). Assuming $\mathbf{d} = d_r(r)\hat{\mathbf{r}}$ with polar symmetry, it reduces to an ODE. In the polar coordinate, it can be expressed as

$$\left(\frac{d^4}{dr^4} + \frac{2}{r} \frac{d^3}{dr^3} - \frac{3}{r^2} \frac{d^2}{dr^2} + \frac{3}{r^3} \frac{d}{dr} - \frac{3}{r^4} + \ell_M^{-4} \right) d_r(r) = 0. \quad (\text{B.1})$$

This equation is fourth order because the original screening equation contains the biharmonic operator. The terms proportional to $1/r$, $1/r^2$, $1/r^3$ and $1/r^4$ are generated by applying the vector Laplacian twice to a radial vector field. To solve this ODE, alternatively, we consider separating the operator composed of biharmonic radial operator term and screening term, i.e., rewriting it into a form of two Bessel equations:

$$(\Delta + i\ell_M^{-2})(\Delta - i\ell_M^{-2})d_r(r) = 0, \quad (\text{B.2})$$

$$\left(\frac{d^2}{dr^2} + \frac{1}{r} \frac{d}{dr} - \frac{1}{r^2} + (i^{1/2}\ell_M^{-1})^2 \right) d_{r,+}(r) = 0, \quad \left(\frac{d^2}{dr^2} + \frac{1}{r} \frac{d}{dr} - \frac{1}{r^2} + (i^{3/2}\ell_M^{-1})^2 \right) d_{r,-}(r) = 0. \quad (\text{B.3})$$

Here, the trick is not to treat the equation as modified Bessel equation but as standard Bessel equation with complex wave number. This factorization is useful because it separates the fourth-order equation into two second-order equations whose solutions are already known. Then their solutions read

$$d_{r,+}(r) = C_1 J_1(e^{i\pi/4}r/\ell_M) + C_2 Y_1(e^{i\pi/4}r/\ell_M), \quad (\text{B.4})$$

$$d_{r,-}(r) = C_3 J_1(e^{3i\pi/4}r/\ell_M) + C_4 Y_1(e^{3i\pi/4}r/\ell_M). \quad (\text{B.5})$$



Therefore, the first kind of representation of general solution in terms of first and second Bessel function (J_1 and Y_1) can be written as

$$d_r(r) = C_1 J_1(e^{i\pi/4} \kappa_M r) + C_2 Y_1(e^{i\pi/4} \kappa_M r) + C_3 J_1(e^{3i\pi/4} \kappa_M r) + C_4 Y_1(e^{3i\pi/4} \kappa_M r), \quad (\text{B.6})$$

where $\kappa_M = 1/\ell_M$. This form is mathematically compact, but the complex arguments make the physical behavior less transparent.

With definitions of modified Bessel functions (I_1 and K_1)

$$I_n(z) = i^{-n} J_n(iz), \quad Y_n(iz) = i^{n+1} I_n(z) - \frac{2}{\pi} i^{-n} K_n(z), \quad (\text{B.7})$$

we obtain the relation between modified Bessel functions and Bessel functions we have derived

$$J_1(e^{i\pi/4} \kappa_M r) = i I_1(e^{-i\pi/4} \kappa_M r), \quad J_1(e^{3i\pi/4} \kappa_M r) = i I_1(e^{i\pi/4} \kappa_M r), \quad (\text{B.8})$$

$$Y_1(e^{i\pi/4} \kappa_M r) = -I_1(e^{-i\pi/4} \kappa_M r) + \frac{2}{\pi} i K_1(e^{-i\pi/4} \kappa_M r), \quad Y_1(e^{3i\pi/4} \kappa_M r) = -I_1(e^{i\pi/4} \kappa_M r) + \frac{2}{\pi} i K_1(e^{i\pi/4} \kappa_M r). \quad (\text{B.9})$$

Therefore, the second kind of representation of general solution in terms of modified Bessel function can be written as

$$d_r(r) = C_1 K_1(e^{i\pi/4} \kappa_M r) + C_2 K_1(e^{-i\pi/4} \kappa_M r) + C_3 I_1(e^{i\pi/4} \kappa_M r) + C_4 I_1(e^{-i\pi/4} \kappa_M r). \quad (\text{B.10})$$

In this representation, the K_1 terms correspond to the decaying sector, while the I_1 terms are associated with the growing sector.

Using definitions of Kelvin functions ($\ker_1$, $\kei_1$, $\ber_1$ and $\bei_1$)

$$\ker_1(z) + i\kei_1(z) = e^{-i\pi/2} K_1(ze^{i\pi/4}), \quad \ber_1(z) + i\bei_1(z) = J_1(ze^{3i\pi/4}) = i I_1(ze^{i\pi/4}), \quad (\text{B.11})$$

and equations representing some properties of modified Bessel functions

$$I_n(ze^{i\pi}) = e^{i\pi n} I_n(z), \quad \overline{I_n(z)} = I_n(\bar{z}), \quad \overline{K_n(z)} = K_n(\bar{z}), \quad (\text{B.12})$$



we can derive the following equations that relate Kelvin functions and modified Bessel functions:

$$\ker_1(z) + i\operatorname{kei}_1(z) = -iK_1(ze^{i\pi/4}),$$

$$\operatorname{ber}_1(z) + i\operatorname{bei}_1(z) = iI_1(ze^{i\pi/4}),$$

$$\ker_1(z) - i\operatorname{kei}_1(z) = iK_1(ze^{-i\pi/4}),$$

$$\operatorname{ber}_1(z) - i\operatorname{bei}_1(z) = -iI_1(ze^{-i\pi/4}),$$

and

$$\ker_1(z) = \frac{1}{2i} \left[K_1(ze^{i\pi/4}) - K_1(ze^{-i\pi/4}) \right],$$

$$\operatorname{kei}_1(z) = -\frac{1}{2} \left[K_1(ze^{i\pi/4}) + K_1(ze^{-i\pi/4}) \right],$$

$$\operatorname{ber}_1(z) = \frac{i}{2} \left[I_1(ze^{i\pi/4}) - I_1(ze^{-i\pi/4}) \right],$$

$$\operatorname{bei}_1(z) = \frac{1}{2} \left[I_1(ze^{i\pi/4}) + I_1(ze^{-i\pi/4}) \right].$$

Therefore, the third kind of representation of general solution in terms of Kelvin functions can be written as

$$d_r(r) = C_1 \ker_1(\kappa_M r) + C_2 \operatorname{kei}_1(\kappa_M r) + C_3 \operatorname{ber}_1(\kappa_M r) + C_4 \operatorname{bei}_1(\kappa_M r) \quad (\text{B.13})$$

The four Kelvin functions have been checked to satisfy Eq. (B.1) using Mathematica. Eq. (B.6), Eq. (B.10) and Eq. (B.13) are three different representations of general solution, where the last representation is what we mainly use, as Eq. (3.4). Note that the Kelvin functions are real functions, and their asymptotic behavior at large r is considered to describe the general solution. This is the main reason for using Kelvin functions in the main text: they keep the solution real while retaining both the oscillatory and decaying features of complex Bessel functions.

B.2 Asymptotic behavior

As $x \rightarrow \infty$, the asymptotic behavior of Kelvin functions $\ker$, $\kei$ are (from [30], which can be verified by Mathematica)

$$\ker_\nu(x) \simeq \sqrt{\frac{\pi}{2x}} e^{-x/\sqrt{2}} \sum_{k=0}^{\infty} \frac{a_k(\nu)}{x^k} \cos\left(\frac{x}{\sqrt{2}} + \left(\frac{\nu}{2} + \frac{k}{4} + \frac{1}{8}\right)\pi\right), \quad (\text{B.14})$$

$$\kei_\nu(x) \simeq -\sqrt{\frac{\pi}{2x}} e^{-x/\sqrt{2}} \sum_{k=0}^{\infty} \frac{a_k(\nu)}{x^k} \sin\left(\frac{x}{\sqrt{2}} + \left(\frac{\nu}{2} + \frac{k}{4} + \frac{1}{8}\right)\pi\right), \quad (\text{B.15})$$

where

$$a_0(\nu) = 1, \quad a_k(\nu) = \frac{(4\nu^2 - 1^2)(4\nu^2 - 3^2) \cdots (4\nu^2 - (2k-1)^2)}{k!8^k}. \quad (\text{B.16})$$

For $\ker_1$ and $\kei_1$, we drop terms with $k \geq 1$ because larger x^k terms decay further faster, and obtain

$$\ker_1(x) \simeq \sqrt{\frac{\pi}{2x}} e^{-x/\sqrt{2}} \cos\left(\frac{x}{\sqrt{2}} + \frac{\pi}{2} + \frac{\pi}{8}\right) = -\sqrt{\frac{\pi}{2x}} e^{-x/\sqrt{2}} \sin\left(\frac{x}{\sqrt{2}} + \frac{\pi}{8}\right), \quad (\text{B.17})$$

$$\kei_1(x) \simeq -\sqrt{\frac{\pi}{2x}} e^{-x/\sqrt{2}} \sin\left(\frac{x}{\sqrt{2}} + \frac{\pi}{2} + \frac{\pi}{8}\right) = -\sqrt{\frac{\pi}{2x}} e^{-x/\sqrt{2}} \cos\left(\frac{x}{\sqrt{2}} + \frac{\pi}{8}\right). \quad (\text{B.18})$$

The leading-order terms are sufficient for identifying the envelope and phase of the screened displacement field. They show explicitly that the decaying Kelvin functions contain both an exponential factor and an oscillatory factor.

For our unique solution Eq. (3.5), its asymptotic behavior at large r reads

$$\begin{aligned} d_r(r) &\simeq d_0 \frac{\sqrt{\frac{\pi}{2\kappa_M r}} e^{-\kappa_M r/\sqrt{2}} \sqrt{\frac{\pi}{2\kappa_M r_{\text{out}}}} e^{-\kappa_M r_{\text{out}}/\sqrt{2}}}{\sqrt{\frac{\pi}{2\kappa_M r_{\text{in}}}} e^{-\kappa_M r_{\text{in}}/\sqrt{2}} \sqrt{\frac{\pi}{2\kappa_M r_{\text{out}}}} e^{-\kappa_M r_{\text{out}}/\sqrt{2}}} \frac{\sin(\frac{\kappa_M r}{\sqrt{2}} + \frac{\pi}{8}) \cos(\frac{\kappa_M r_{\text{out}}}{\sqrt{2}} + \frac{\pi}{8}) - \cos(\frac{\kappa_M r}{\sqrt{2}} + \frac{\pi}{8}) \sin(\frac{\kappa_M r_{\text{out}}}{\sqrt{2}} + \frac{\pi}{8})}{\sin(\frac{\kappa_M r_{\text{in}}}{\sqrt{2}} + \frac{\pi}{8}) \cos(\frac{\kappa_M r_{\text{out}}}{\sqrt{2}} + \frac{\pi}{8}) - \cos(\frac{\kappa_M r_{\text{in}}}{\sqrt{2}} + \frac{\pi}{8}) \sin(\frac{\kappa_M r_{\text{out}}}{\sqrt{2}} + \frac{\pi}{8})} \\ &= d_0 \sqrt{\frac{r_{\text{in}}}{r}} e^{-\kappa_M(r-r_{\text{in}})/\sqrt{2}} \frac{\frac{1}{2} \left[\sin(\frac{\kappa_M(r+r_{\text{out}})}{\sqrt{2}} + \frac{\pi}{4}) + \sin(\frac{\kappa_M(r-r_{\text{out}})}{\sqrt{2}}) \right] - \frac{1}{2} \left[\sin(\frac{\kappa_M(r+r_{\text{out}})}{\sqrt{2}} + \frac{\pi}{4}) + \sin(\frac{\kappa_M(r_{\text{out}}-r)}{\sqrt{2}}) \right]}{\frac{1}{2} \left[\sin(\frac{\kappa_M(r_{\text{in}}+r_{\text{out}})}{\sqrt{2}} + \frac{\pi}{4}) + \sin(\frac{\kappa_M(r_{\text{in}}-r_{\text{out}})}{\sqrt{2}}) \right] - \frac{1}{2} \left[\sin(\frac{\kappa_M(r_{\text{in}}+r_{\text{out}})}{\sqrt{2}} + \frac{\pi}{4}) + \sin(\frac{\kappa_M(r_{\text{out}}-r_{\text{in}})}{\sqrt{2}}) \right]} \\ &= d_0 \sqrt{\frac{r_{\text{in}}}{r}} e^{-\kappa_M(r-r_{\text{in}})/\sqrt{2}} \frac{\sin(\frac{\kappa_M(r-r_{\text{out}})}{\sqrt{2}})}{\sin(\frac{\kappa_M(r_{\text{in}}-r_{\text{out}})}{\sqrt{2}})} \\ &\sim \frac{1}{\sqrt{r}} e^{-\kappa_M r/\sqrt{2}} \sin\left(\frac{\kappa_M}{\sqrt{2}}(r_{\text{out}} - r)\right) \end{aligned} \quad (\text{B.19})$$



where the last line Eq. (B.19) is the asymptotic solution. The factor $e^{-\kappa_M r / \sqrt{2}}$ gives the screening envelope, and the factor $1/\sqrt{r}$ is the usual radial spreading contribution in two-dimensional polar geometry. This is why the main text defines the monopole screening length as $L_M = \sqrt{2}\ell_M$.

B.3 Oscillations of the particular solution

To find the key characteristics of oscillations in the particular solution Eq. (3.5), we start with the asymptotic expansion Eq. (B.19):

$$\begin{aligned}
 d_r(r) &\sim \frac{1}{\sqrt{r}} e^{-\kappa_M r / \sqrt{2}} \frac{\sin(\frac{\kappa_M}{\sqrt{2}}(r_{\text{out}} - r))}{\sin(\frac{\kappa_M}{\sqrt{2}}r_{\text{out}})} \simeq \frac{1}{\sqrt{r}} e^{-\kappa_M r / \sqrt{2}} \frac{\sin(n\pi - \frac{\kappa_M}{\sqrt{2}}r)}{\sin(n\pi + o^\pm)} \\
 &\simeq \frac{-1}{\sqrt{r}} e^{-\kappa_M r / \sqrt{2}} \frac{\sin(\frac{\kappa_M}{\sqrt{2}}r)}{\sin(o^\pm)} = -\frac{1}{\sin(o^\pm)} \frac{\sin(\kappa_M r / \sqrt{2})}{\sqrt{r}} e^{-\kappa_M r / \sqrt{2}} \\
 &\simeq -\frac{1}{\sin(o^\pm)} \frac{\sin(n\pi r / r_{\text{out}})}{\sqrt{r}} e^{-\kappa_M r / \sqrt{2}} \\
 &\sim -\frac{1}{\sin(o^\pm)} \frac{\sin(n\pi \hat{r})}{\sqrt{\hat{r}}} (e^{-\kappa_M \hat{r} / \sqrt{2}})^{r_{\text{out}}}
 \end{aligned} \tag{B.20}$$

where $o^\pm \equiv \frac{\kappa_M}{\sqrt{2}}r_{\text{out}} - n\pi \rightarrow 0^\pm$ when $\kappa_M \simeq \sqrt{2}n\pi/r_{\text{out}}$. So the asymptotic expansion of $d_r(r)$ at large r is $e^{-\kappa_M r / \sqrt{2}} \sin(n\pi r / r_{\text{out}}) / \sqrt{r}$ as $\kappa_M \rightarrow (n\pi)^-$ and $-e^{-\kappa_M r / \sqrt{2}} \sin(n\pi r / r_{\text{out}}) / \sqrt{r}$ as $\kappa_M \rightarrow (n\pi)^+$ where $r/r_{\text{out}} \in (0, 1]$. Fig. 3.1 can be explained by Eq. (B.20). The sign change across the critical value reflects the small denominator in the boundary-value solution. Thus the oscillatory profiles shown in the figures are not accidental numerical features, but direct consequences of the finite outer boundary and the Kelvin-function structure of the solution.



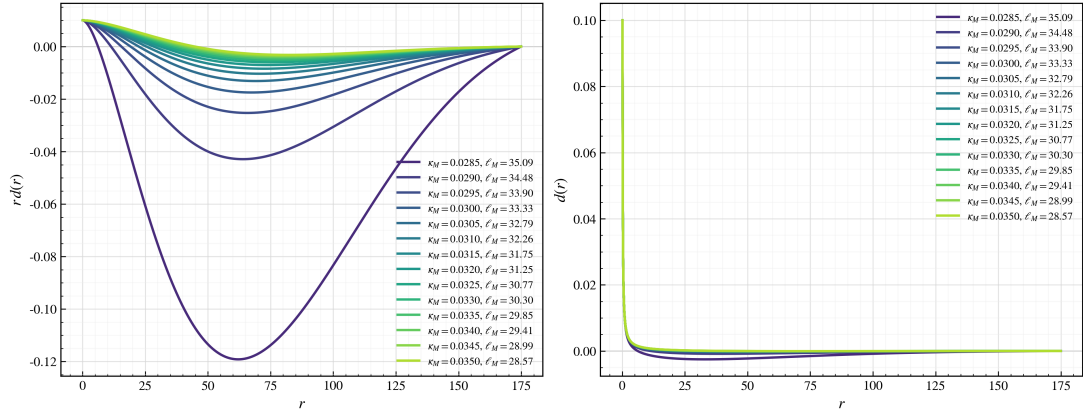
Appendix C Supplementary Figures

C.1 Math properties of the particular monopole screening solution

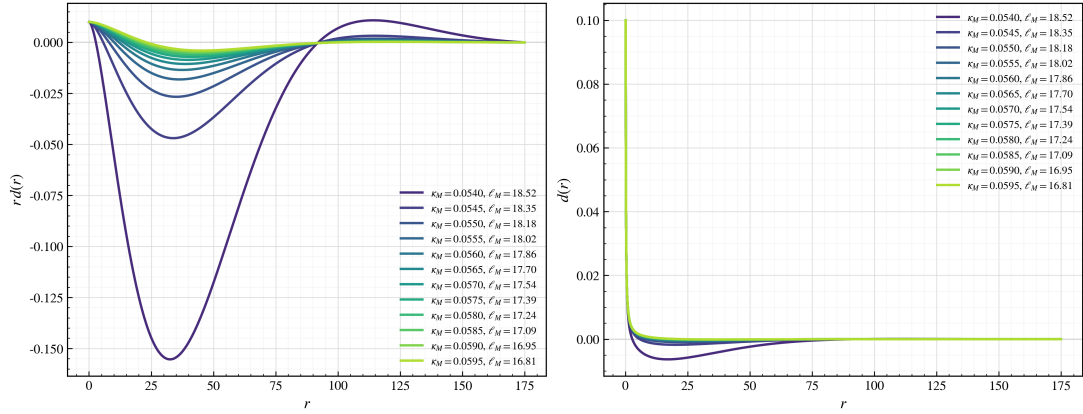
The rest graphs Fig. C.1, Fig. C.2 and Fig. C.3 for details are in this section, whose plot settings are the same as Fig. 3.1, Fig. 3.2 and Fig. 3.3 in the main context. In particular, the “ $-\sin$ ” samples are shown in Fig. C.1 while the “ $+\sin$ ” samples are shown in Fig. 3.1. Together, these figures display the two possible signs of the oscillatory branch near the characteristic values of κ_M . They supplement the main-text figures by showing that the qualitative behavior is robust across the selected wavenumber ranges.

C.2 Fitting results

The rest graphs Fig. C.4 and Fig. C.5 for details are in this section, whose plot settings are the same as Fig. 4.1 and Fig. 4.2 in the main context. All wave numbers fitted from 195 samples are shown in Figure C.6, where the dispersion of data points shows the huge fluctuations between samples intuitively. For correlation length scale ξ , there is almost no difference between the fitting-first and averaged-first results. The former is shown in Fig. C.7 without error bars and the latter is shown in Fig. 4.8. These supplementary figures are included to make the fitting procedure more transparent. The main text focuses on representative panels, whereas the appendix records the remaining packing fractions and the full distribution of fitted wave numbers.



(a) $-\sin, L/\lambda \sim 1/2$



(b) $-\sin, L/\lambda \sim 1$

FIG. C.1 Solution curves of Eq. (3.5) for different κ_M .

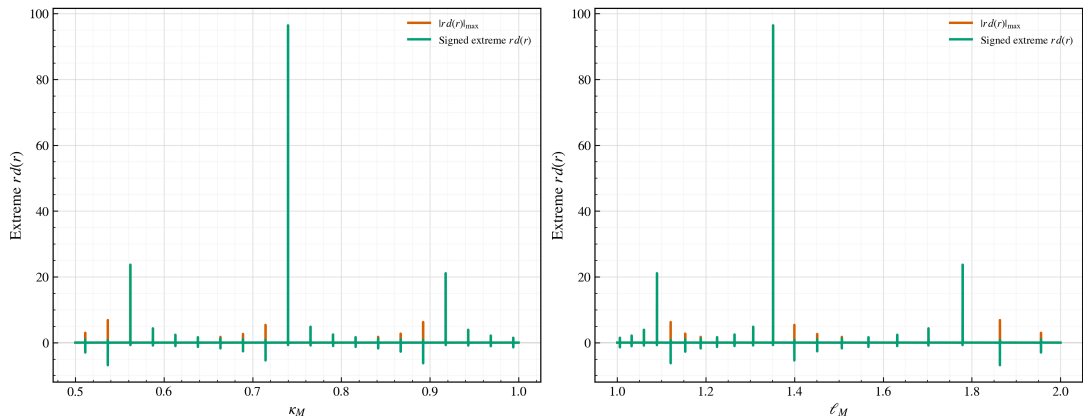
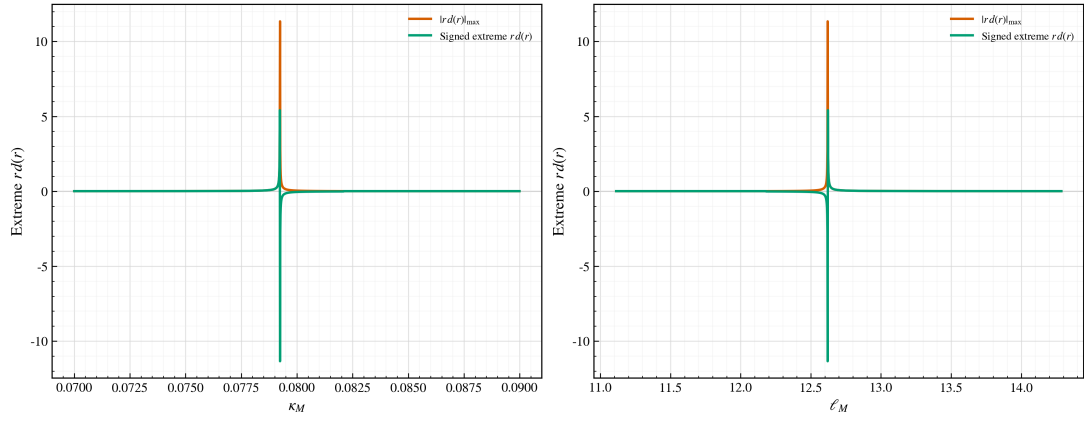
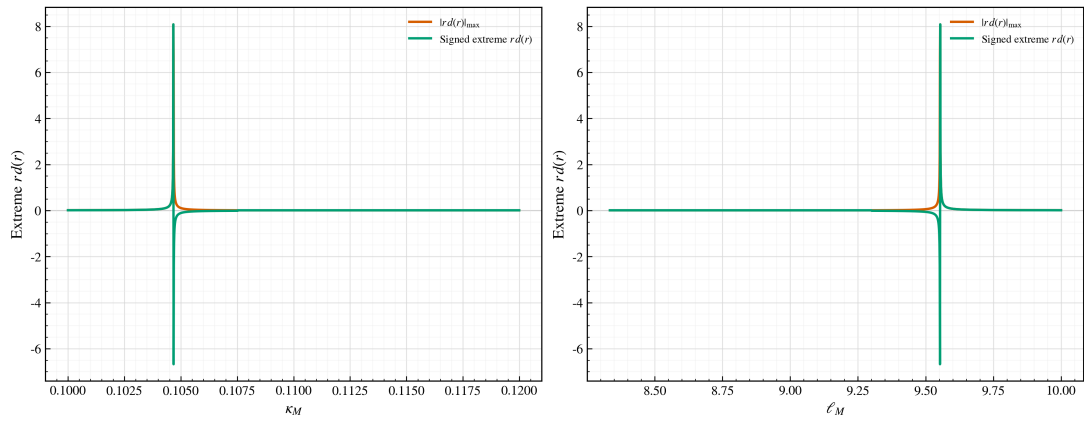


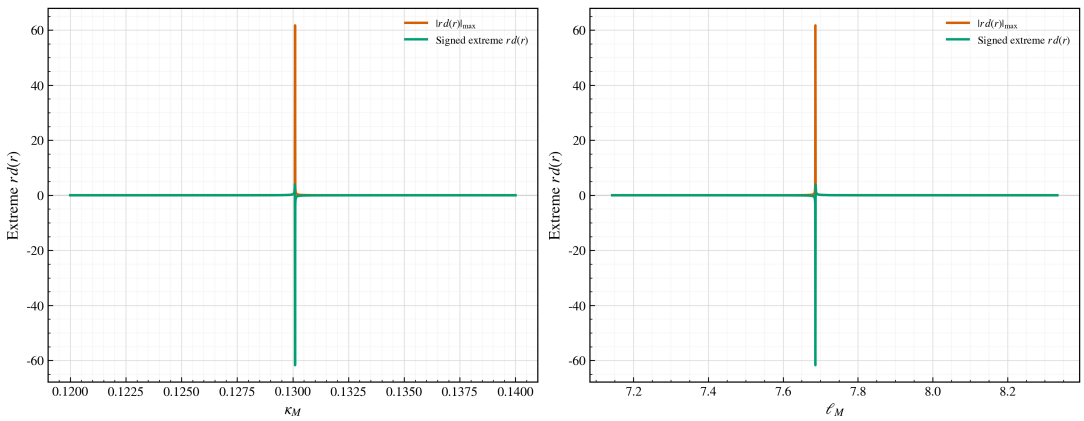
FIG. C.2 Maximum $|rd_r|$ and rd_r plotted against κ_M (left) and ℓ_M (right). κ_M in (0.5, 1.0), ℓ_M in (1, 2).



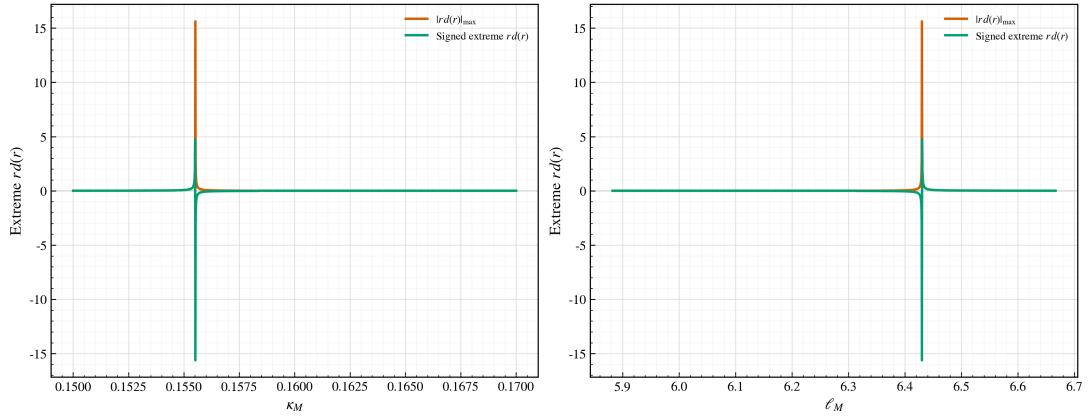
(a) Critical $\kappa_M = 0.07924$ and $\ell_M = 12.62$



(b) Critical $\kappa_M = 0.10467$ and $\ell_M = 9.55$

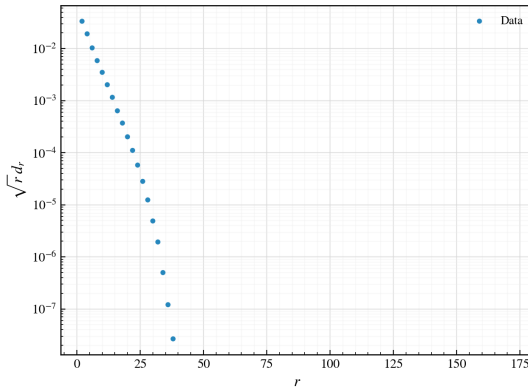


(c) Critical $\kappa_M = 0.13010$ and $\ell_M = 7.69$

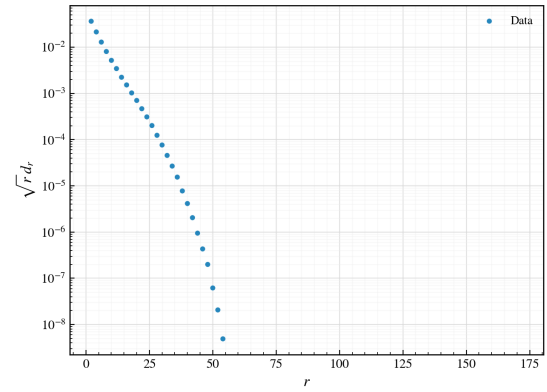


(d) Critical $\kappa_M = 0.15552$ and $\ell_M = 6.43$

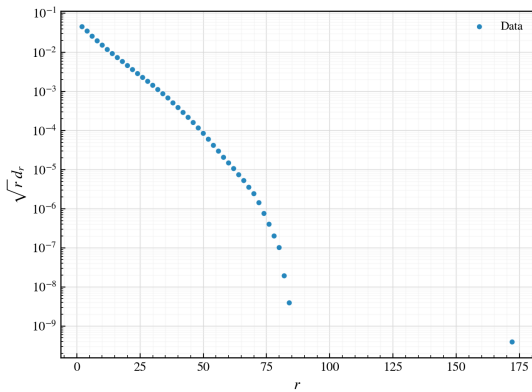
FIG. C.3 Critical values of κ_M and ℓ_M for different wavenumber regimes.



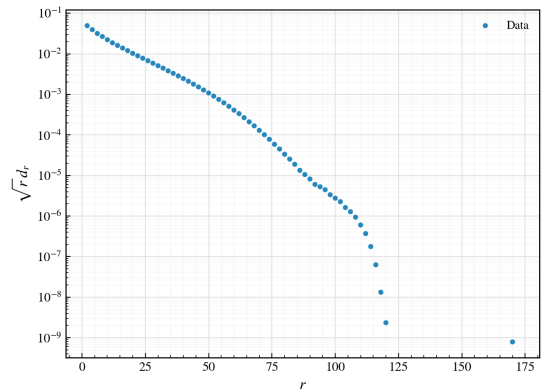
(a) $\phi = 0.7301$



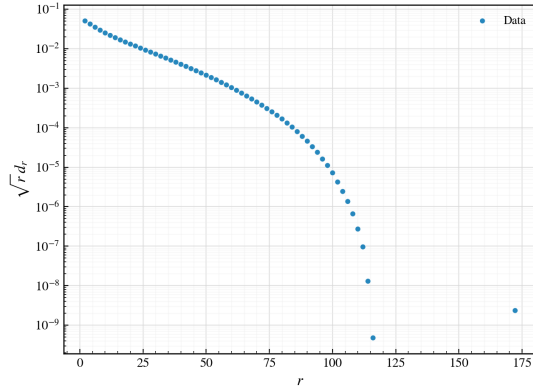
(b) $\phi = 0.7602$



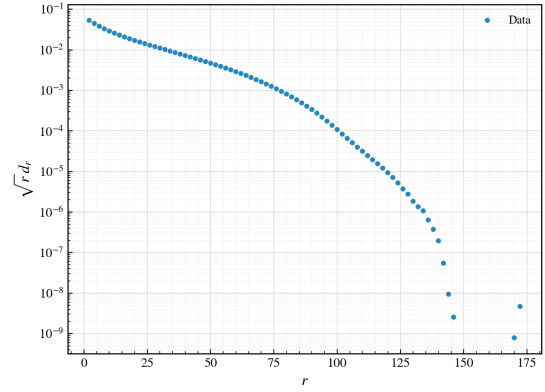
(c) $\phi = 0.7926$



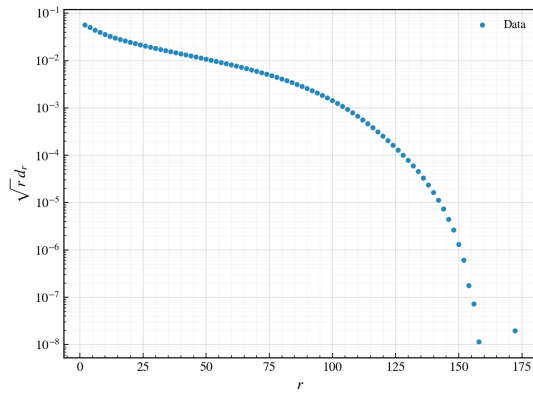
(d) $\phi = 0.8090$



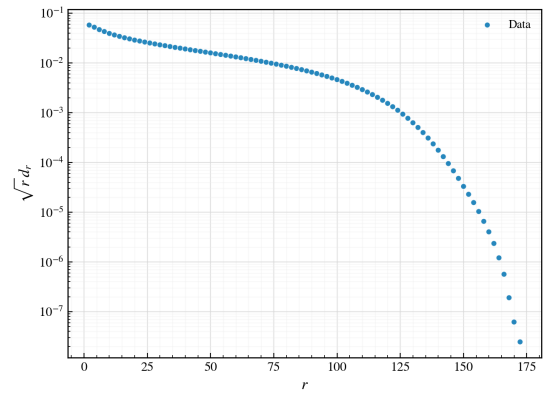
(e) $\phi = 0.8173$



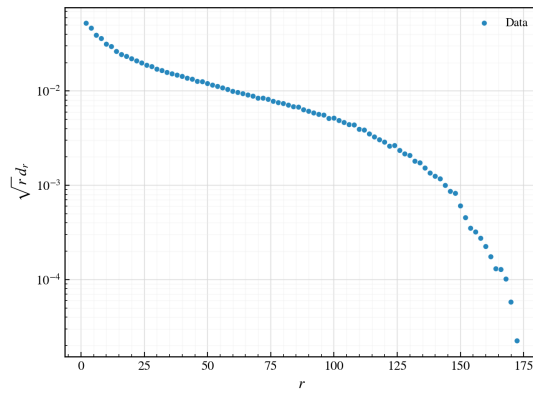
(f) $\phi = 0.8256$



(g) $\phi = 0.8339$

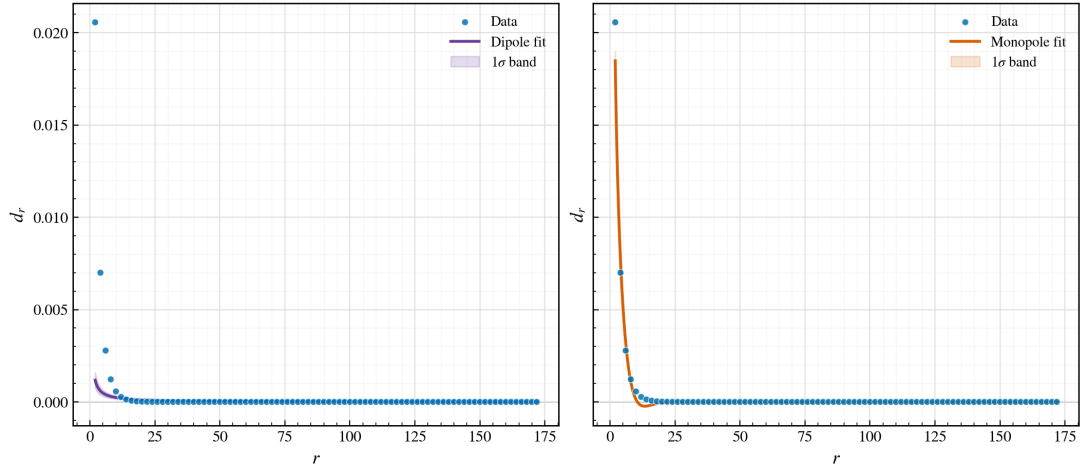


(h) $\phi = 0.8390$

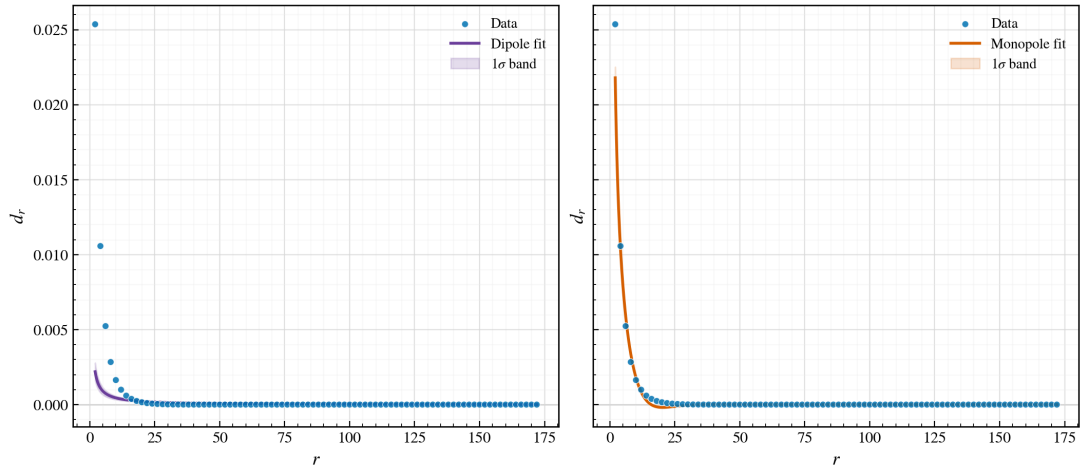


(i) $\phi = 0.8410$

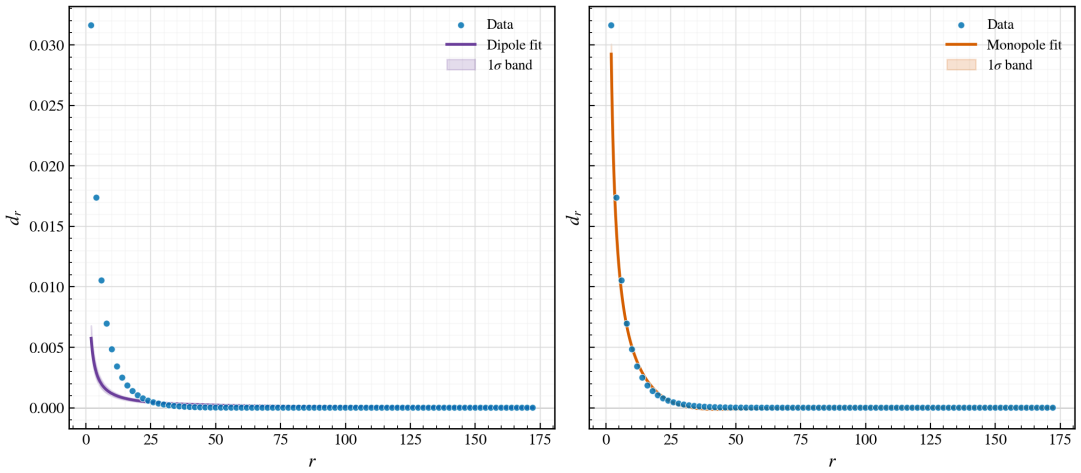
FIG. C.4 Log-linear plots of the averaged scaled radial displacement field $\sqrt{r}d_r$.



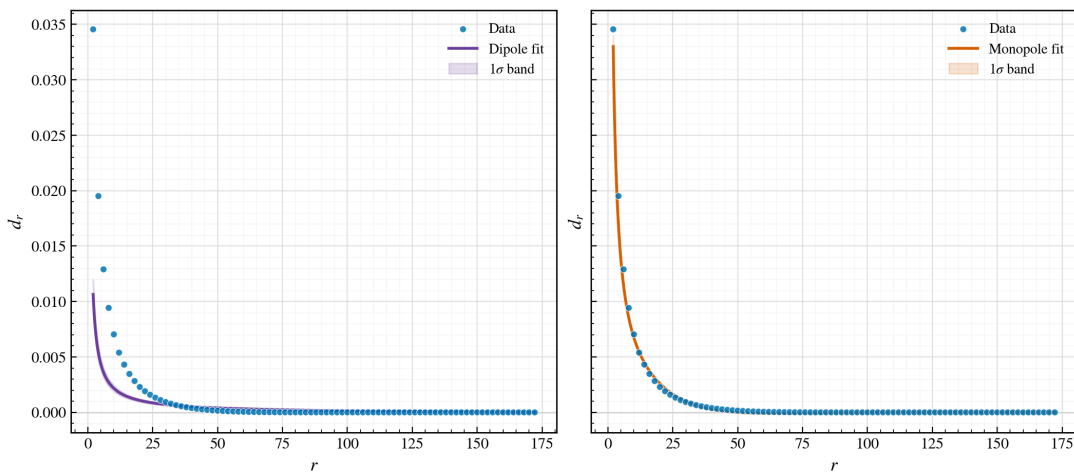
(a) $\phi = 0.7129$: $\ell_M = 2.809$



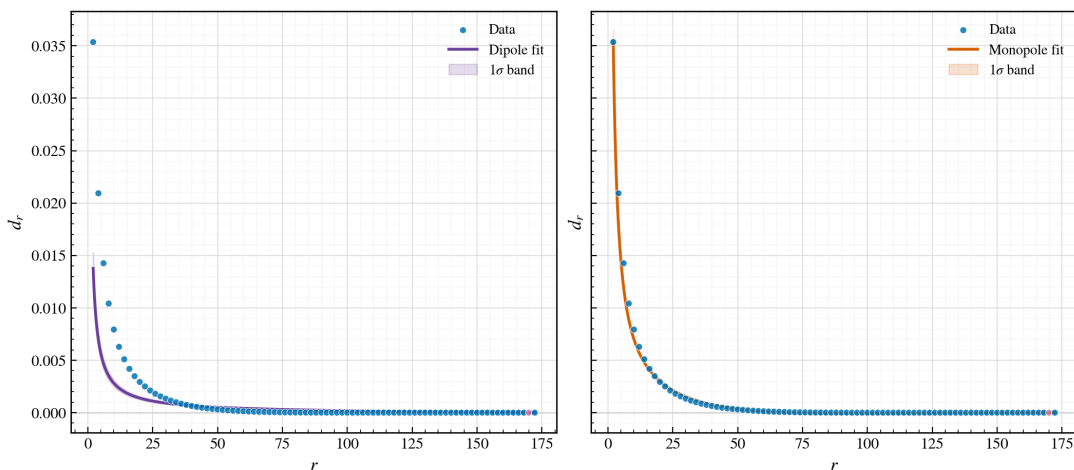
(b) $\phi = 0.7602$: $\ell_M = 4.411$



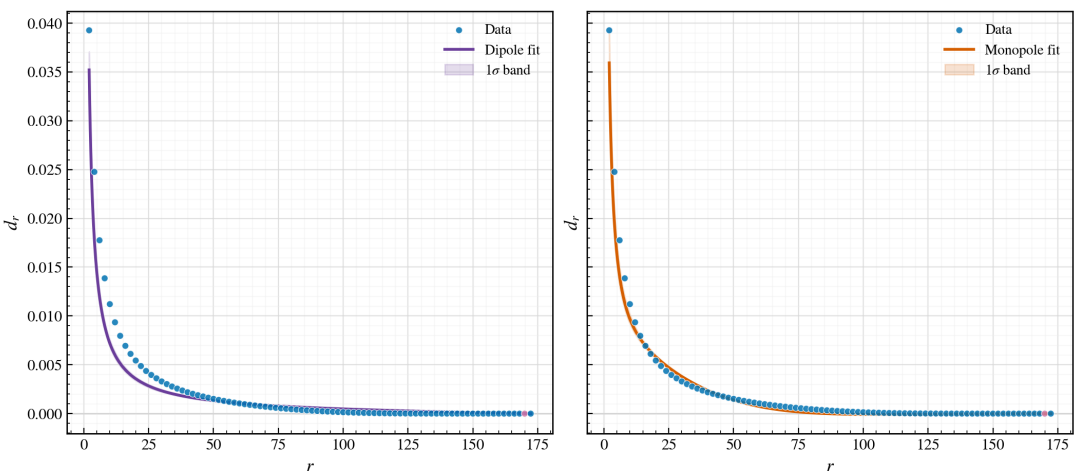
(c) $\phi = 0.7926$: $\ell_M = 10.436$



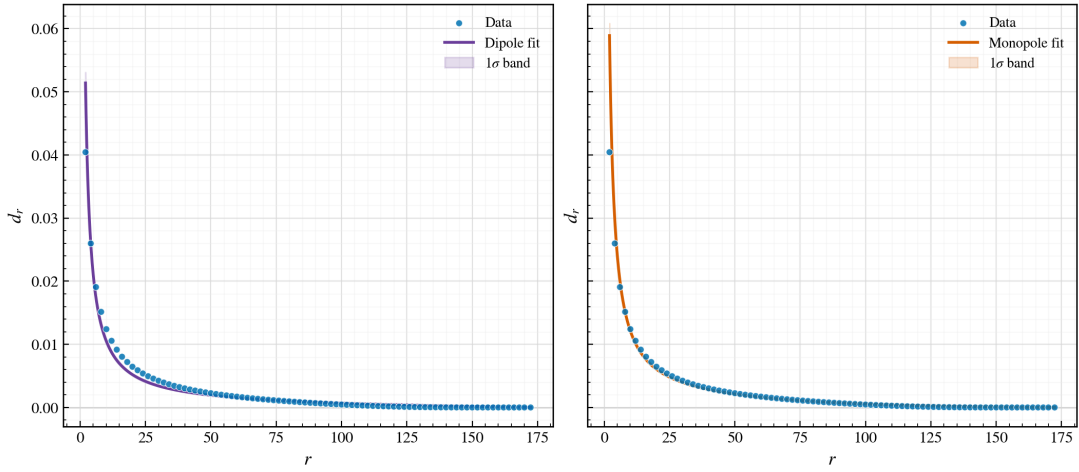
(d) $\phi = 0.8090$: $\ell_M = 13.833$



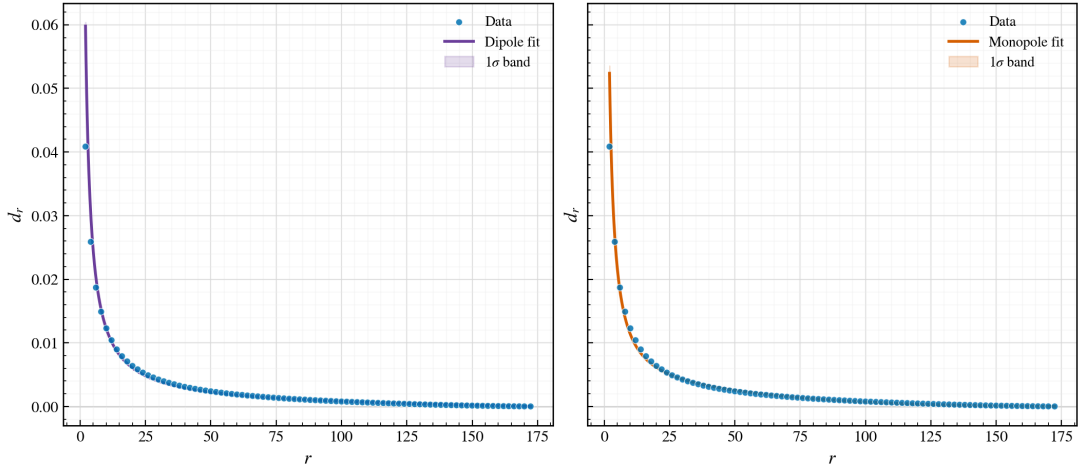
(e) $\phi = 0.8173$: $\ell_M = 23.149$



(f) $\phi = 0.8339$: $\ell_M = 19.456$



(g) $\phi = 0.8390$: $\ell_M = 52.048$



(h) $\phi = 0.8407$: $\ell_M = 42.469$

FIG. C.5 Left panel: fitting results of dipole screening. Right panel: fitting results of monopole screening.

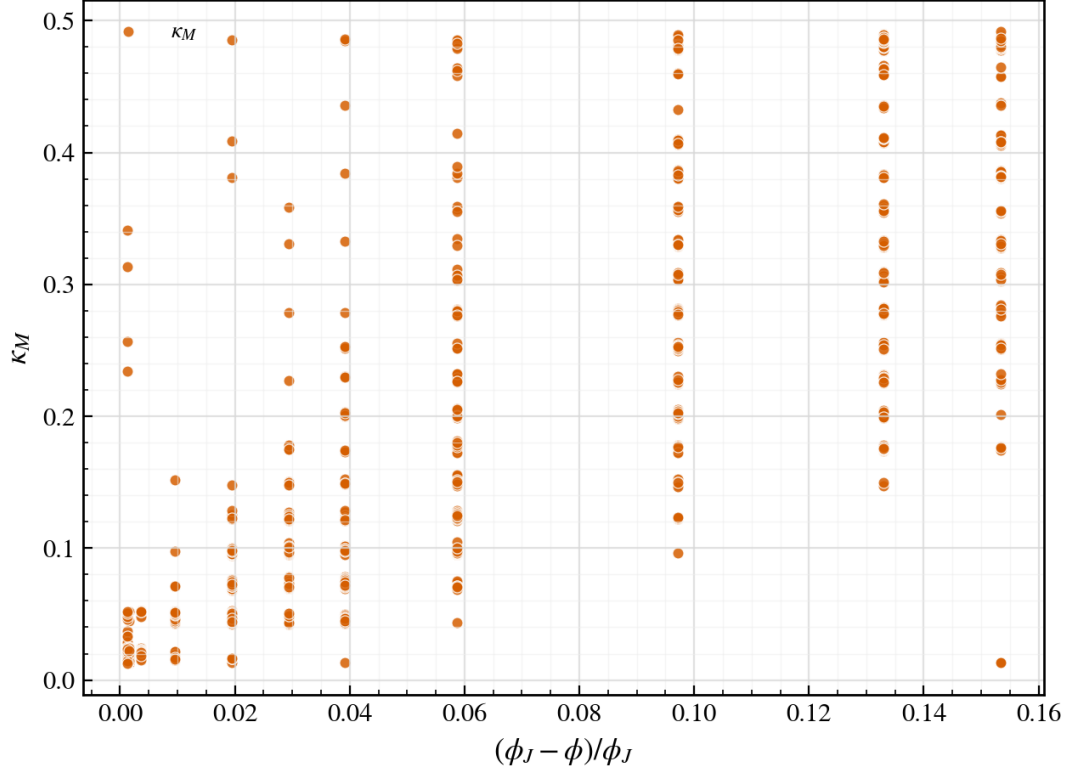


FIG. C.6 All κ_M points versus $(\phi_J - \phi)/\phi_J$.

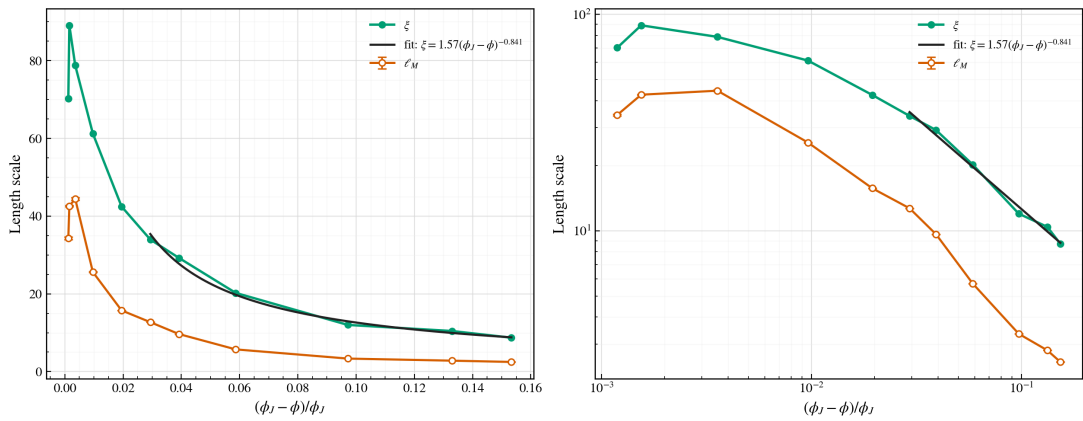


FIG. C.7 Length scales ξ and ℓ_M as functions of $(\phi_J - \phi)/\phi_J$ in linear (left panel) and log-log (right panel) plot. Here, we obtain ξ with fitting the averaged data.